

Calculus I: Notes and Problems

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To the Instructor

This problem sequence is intended for a first semester course in calculus for STEM (science, technology, engineering, and mathematics) majors. The course is often designated as calculus I. Where I teach, the course is supposed to cover differentiation, limits, continuity, and the fundamental theorem of calculus. We do not cover applications of the definite integral or techniques of integration beyond simple antidifferentiation rules and running the chain rule backwards, something often referred to as u-substitution. I have used some version of these notes for 9 semesters starting with fall 2002. I owe a great deal to and have borrowed much from the late Dr. William S. Mahavier who taught mathematics in the spirit of R.L. Moore for roughly 50 years and whose teaching I had the great fortune to observe during an entire semester at Emory University during the fall of 2001.

Calculus is a particularly difficult course in which to implement inquiry-based or Moore method approaches since it is often perceived to be a course in which the student becomes proficient at executing a number of standard algorithms and then applying them in a variety of mostly standard situations. Many students will proceed to a course immediately after calculus I that is typically called calculus II. The perception is that if one fails to "cover" the material on the syllabus in calculus I, then the students will not be prepared to take calculus II. The Moore method approach requires that students be given adequate time to digest the principles that underlie the techniques, that they are able to generate justifications on their own, and that they are able to communicate their ideas clearly and precisely. If one gives honest priority to these goals it is often at the expense of covering all the "required" topics. Some topics must therefore be omitted. It has been my experience, however, that the depth versus breadth approach leaves the students sufficiently prepared for calculus II.

When using these notes the question of how quickly a class should proceed through the problem sequence is still a legitimate concern. In any IBL course the instructor should expect the course to move much slower at the beginning than at the end. It will take most of the students some time to get used to this learning style. They will spend a considerable amount of time struggling with how to approach problems for which a solution method has not been prescribed. They will also be struggling with how to explain why

a particular solution is correct and how to present their solutions with the right amount of detail and clarity to satisfy an audience of fellow students. During a good semester which for me consists of 15 weeks of class with four 50 minute sessions per week the pace at which we go through the sequence is very roughly as follows. Chapter 1 takes about 4 weeks, chapter 2 about 7 weeks, and chapter 3 about 4 weeks.

Just getting comfortable presenting mathematical material to any audience is quite a challenge. Getting students to progress through these initial challenges can require considerable time and cannot be forced. Pushing forward when the majority of the students are not keeping up will likely result in much slower progress later on. Nothing will be gained. On the other hand, careful attention to promoting good habits at the beginning can have huge benefits in improving the pace of the course toward the end.

There are, however, still reasons to be concerned about pacing. It can happen, and I know this from personal experience, that a class is moving slowly simply because students are not putting forth enough effort. For example, when the deadlines and the grading structure are not as well defined as in some other classes, students might be tempted to put working on a challenging problem in calculus on the back burner. There are carrot and stick approaches to combat this problem. You will have to see what works best for you, but it deserves attention because a class in which day after day goes by without any student presentations can make you wish you were lecturing instead. Then at least you would feel like you were making progress.

The best carrot is getting a critical mass of students fired up about the course. The satisfaction they get from successfully working problems can easily infect the rest of the class. You can encourage this by showing very openly your own great satisfaction with student accomplishments. Most of the time this will be genuine. Even if it is not, it can't hurt. I encourage students to clap when a good presentation has been completed. Often this is completely spontaneous, which is even better. Sometimes when a particular problem has gone unsolved for a long time I will increase the point value associated with it as an extra incentive. If nothing else, this brings extra prestige to the student that ultimately solves the problem.

If none of this works and you must bring out the stick, then here are some things I have tried. First of all, at the end of a given class session I will tell the students which problems they are expected to have ready for the next class period. Usually this is done by way of email or some other electronic posting immediately after class. When the students come to class the next time, they know that they might be called on to present any of the problems that I have posted. When selecting students to present I either select the students at random or select the student that has presented the least first. The student called on must either present what they have or confess to not having anything. Usually, if they have anything at all they are encouraged

to show what they have and to explain what, if anything, kept them from finishing the problem. If they have nothing then I will typically ask why not. If “I just didn’t get to it” is the response too often then I will encourage an office visit during which I try to get the student back on track. If I have a whole class of students who are putting forth little effort I will often give a written assignment that covers problems that have not yet been done in class with a deadline in the near future. In order to have the best chance of successfully completing this assignment the students will need to see the problems worked in class with feedback. At other times when no one has anything I will ask everyone to write down the problem numbers that they were expected to have ready for the particular class period and beside each number indicate what they tried, what specifically kept them from making progress, and what if anything might help them solve the problem for the next class period.

It can happen that even though you have done all that you can to keep the class moving at a reasonable pace it will be impossible to finish all the problems in the problem sequence. There are certain things that can be left out without disturbing the flow of the course. For example if you have not made it through section 2.2 by the middle of the semester then you might consider just stating the derivative of the sine function instead of having the students derive it. That means you would skip problems 89 through 101. If you are very far behind you can skip all the derivations of trig function derivatives and skip problems 89 through 106. You can skip entirely the optimization problems in section 2.7 or just do some subset of them without any interruption of the flow. You can also skip the derivation of the derivatives of the inverse trig functions if absolutely necessary. If the end of the semester is fast approaching (you only have a couple of weeks left) and you have not gotten to problem 176 you can consider skipping 176, 177, and 181-185 which are just two examples of situations where exponential functions arise as solutions to initial value problems.

For more on teaching using a Moore Method approach see *The Moore Method, A Pathway to Learner-Centered Instruction*, by Charles A. Coppin, W. Ted Mahavier, E. Lee May, and G. Edgar Parker.

I have two major goals for my calculus I students. One, I want my students to understand the big ideas behind calculus. And two, I want to get my students to understand on their own terms how and why the techniques that are used in calculus work and I want them to be able to articulate this understanding in a clear and precise way.

There are many side benefits that students typically gain from taking a Moore method class and they should not be taken lightly. These include confidence, persistence, independence, creativity, communication skills, and reduced fear of public speaking. Often a student is transformed from being a passive learner to someone who takes an active role in their own learning. If

you are convinced that these benefits outweigh the potential risks associated with Moore method teaching then you should consider using these notes.

You will notice that I do not use certain notational conventions common to standard calculus textbooks. I do this in order to give highest priority to allowing the student to make sense of the mathematics. I try to stay away from any symbolic conventions that have the potential to mislead the student or that encourage mindless symbol manipulation. If the symbols are not clear and transparent then confidence in the validity of the manipulations is easily eroded. If a symbol manipulation has been performed that is not transparent then a student will have great difficulty justifying why the procedure works. A guiding principle to any inquiry based approach is that the student should at any point in the course be able to justify their work. When notation works against this guiding principle I try to avoid it. I find that it is much easier for a student to go from a firm understanding of the mathematics to dealing with confusing notation than the other way around. Once a technique has been learned as simply a notational trick, it is hard for the student to go back and try to justify it, especially if the notation is misleading.

For more on this see for example *PRIMUS*, XVIII(4): 349–360, 2008 Calculus: The Importance of Precise Notation, William S. Mahavier and W. Ted Mahavier.

You will also notice that limits and continuity are de-emphasized, with many details relegated to an appendix. This approach, although somewhat unusual, is not out of line with many textbooks that came out of the reform calculus movement. See for example Hughes-Hallet or Ostebee and Zorn or much earlier textbooks such as Lange's First Course in Calculus. Certainly the discoverers of calculus Newton and Leibnitz were content with only an intuitive understanding of these concepts.

In the attempt to construct an approach to calculus in which it is the students that provide the justifications for the rules and procedures that are followed I had to make sure that those justifications were at their level.

Even though most students in a first semester can learn to mimic an epsilon delta proof for a limit of a linear function, it has been my experience that only a handful of students at that level have any real understanding of what they are doing and almost no chance of creating correct proofs of this nature absent a template.

This is not to say that such justification is unimportant or should be ignored entirely. Certainly in a course at the level of real analysis or topology one would expect students to construct, on their own, correct proofs of limit theorems, and continuity statements.

Why not, then, just treat the differentiation rules as axioms as well and move on? My feeling is that these rules can be proved by students at the

first year calculus level just as a student in elementary school can provide a justification of how to add fractions. A proof constructed by a student can provide the confidence that is necessary in order for the student to independently recreate a formula or determine what can be generalized. Is it true that the derivative of a product is the product of the derivatives? Is the derivative of a^x xa^{x-1} ? These are questions that the student who has constructed proofs of the derivative rules has a chance of being able to answer and the student that has simply been asked to memorize the rules would struggle to answer without outside help.

The items in the appendix are there so as not to distract from the flow of the problem sequence. An astute student may be interested in the limit theorems and wish to prove some of them. That student should be directed to the appendix where they can work through the problems on their own. For everyone else the theorems are meant to be accepted as axioms and their formal justification is left to a later time. The introduction of the limit rules should be accompanied by a classroom discussion during which the students hopefully gain some intuition regarding the rules. How deep this discussion gets will depend on the interest and mathematical sophistication of the class and can run the full gamut from simple closeness arguments to just short of epsilon delta proofs, but in my experience usually stays closer to the former.

I am concerned that students experience proof as a form of sense-making rather than as a formal string of statements constructed using obscure rules and symbols. It is for this reason that I shy away from proofs that are beyond the reach of the students that I typically encounter in a first semester calculus class.

Although I occasionally hear from other professors that some student who took my calculus class and then went on to take a follow up class reported that “we didn’t do that in Stallmann’s class”, these comments are always about some algorithm they didn’t learn and never about conceptual matters. In particular, a lack of emphasis on continuity in my class has never been raised as an issue. Continuity is not really a topic in our Calculus II class and when it finally reemerges in multivariable calculus it appears in a much different form and the students for the most part have completely forgotten their first exposure to it. As far as I know, students who have seen the definition of continuity in calculus I are no more or less able to conceptualize this new definition in multivariable calculus. I also teach undergraduate topology in which continuity is, of course, a fundamental concept. I recently asked my topology class, none of whom had had me for calculus, whether anyone could remember a definition of continuity or could give me at least a conceptual idea of continuity. No one was able to do this. The best I could get was something about drawing the graph without lifting the pencil, something which translates very poorly to higher dimensional versions of

continuity. All of these students had taken calculus I and II, and most had taken calculus III. Although I don't claim that this is a representative sample of students who have successfully completed a calculus I and II sequence, it does give some credibility to the notion that just because some topic is "covered" in class does not mean that it has any long term staying power.

With a course at this level I find it especially important that the first few problems are easy enough that almost everybody in the class has a chance at getting one of them. High doses of praise are also particularly important at the beginning.

Students should be frequently reminded that the problems are in sequence and that they can and must use the statements, results, and techniques developed in earlier problems in order to be able to tackle the new ones. One of the best strategies for students to follow when they are lost is to review carefully the problems that have gone before. Even if a problem has not been done yet, I allow it to be used in problems that follow it in the sequence.

I usually give the students just enough of the notes to keep them busy until the next few class periods. This allows them to make discoveries on their own that might be revealed in later problems. I frequently add problems that come from questions that the students have asked in class or that reinforce ideas that I believe were poorly grasped. I encourage anyone using these notes to add or subtract as needed to suit the needs of their students.

I typically give two "gateway" tests to check that the students have the routine computational skills that might be required of them in a subsequent course. Students that don't pass the test with at least 70% correct are given another chance but they must set aside time outside of class for this. Usually the test presents no major obstacle to those students who are otherwise keeping up with the course. The first test should be given at roughly the middle of the semester and the second a week or two before the end of the semester. Examples of each are given at the end of this introduction.

I grade students on a 3 point scale for work that they do at the board. The criteria I use are:

- 0-You really have nothing. You are just winging it.
- 1-You have major errors.
- 2-You have some minor errors, your solution is not complete, or your presentation is not clear.
- 3-You have a clear, correct, and complete solution.

Students can redeem a low initial grade by finishing or redoing an earlier attempt.

The more points a student accumulates the better. A student is never penalized for trying. Which number of points corresponds to a particular

letter grade will depend on the number of students in the class, the number of problems you are able to get through in a semester, and to some extent the diligence and talent of your students. I tell my students that I can give them a rough letter grade equivalent anytime they ask after the first few weeks and usually I have no trouble producing a plausible letter grade.

Students turn in written assignments once a week. These assignments are carefully written solutions to problems that have been presented in class, with the stipulation that they are to be accompanied by written explanations in complete sentences. I typically allow students to choose from a list of problems that have been presented since the last written assignment was due. I generally exclude problems from the list that are trivial once the students have seen the solution and those problems which require only routine computations. I use the following 5 point scale for written assignments.

0-The work you handed in is not your own.

1-The work you handed in does not make sense or is irrelevant to the problem.

2-The work you handed in contains major errors that cannot be fixed without a complete overhaul.

3-You have the right idea but the details are incorrect or incomplete.

4-Your solution is essentially correct. What is keeping you from earning a 5 is either poor or imprecise wording or notation or minor computation or algebra errors.

5- Your solution is both correct and written in a clear and precise manner.

The final grade is one third board work, one third written work, and one third final exam. The gateway tests enter the grading scale only if a student does not pass after three attempts. This has never happened to any student that I thought would otherwise have passed. At my institution I can give the first gateway test before the withdrawal deadline, so that a student has the option of withdrawing from the course rather than receiving an F if that student fails the gateway way test three times.

I find that students at the calculus level need to be convinced early of the benefits of the inquiry based approach and reminded frequently throughout the semester.

On the first day of class I facilitate an exercise that proceeds as follows. The students are put in groups of three or four students. They are asked to think of something that they believe they are particularly good at and then to discuss how it was that this was learned. After the groups share their observations with the rest of the class I ask the students to think about something that they were taught, but have very little recollection of. The discussion leads very naturally to a realization of the benefits of an approach that focuses on what the students do rather than what the teacher tells.

I use a camera to assist my students and me in keeping track of what happened in class. I have students write their names by the problem that they are presenting and when they are finished, I take a picture of their work. The pictures get posted to the web within a few hours after class. This allows the students to concentrate on the discussion rather than having to write down everything on the board.

After a section has been completed it is often useful to summarize what has been accomplished using the students work as a guide, (e.g. Mr. A showed us that ... and Ms. B. then demonstrated that ...).

Gateway Test Examples:

Calculus I Gateway Test 1

Instructions. In each of the following the formula for a function f is given. Use the rules for taking derivatives to find a formula for the function f' .

1. $f(x) = 9x^7 + \sqrt{x} - \frac{5}{x}$
2. $f(x) = ax^3 + \frac{x^2}{b} + \frac{x}{c} + d$
3. $f(x) = \sqrt{x^5 - 3x}$
4. $f(x) = \frac{3x}{x^2 + 1}$
5. $f(x) = (2x^3 + 3x)^{12}$
6. $f(x) = (5x^2 + 2x - 1)^6 (x^3 + x)^5$
7. $f(x) = \left(\frac{3x+1}{6-x} \right)^3$
8. $f(x) = \frac{x}{\sqrt{5+x^2}}$
9. $f(x) = (x^3 + 2x - 1) \sqrt{x+1}$
10. $f(x) = \frac{\sqrt{x} + \sqrt{x}}{x}$

Calculus I Gateway Test 2

Part 1 Instructions. In each of the following the formula for a function f is given. Use any rules for taking derivatives to find a formula for the function f' .

1. $f(x) = \sin(x) \exp(x)$
2. $f(x) = \exp(x^3)$
3. $f(x) = \ln(\cos(x))$

Part 2 Instructions. In each of the following the formula for a function f is given. You are to write down the most general form of an antiderivative of that function. Remember to check that you have the correct antiderivative by taking the derivative of the function that you find.

1. $f(x) = \frac{3}{x} - 3 \sin x + 4x^{1/5}$
2. $f(x) = x^2 \exp(x^3)$
3. $f(x) = \frac{\sin x}{2 + \cos x}$
4. $f(x) = \frac{1}{\sqrt{9 - x^2}}$

Part 3 Instructions. In each of the following you are to evaluate the definite integral by using the Fundamental Theorem of Calculus.

1. $\int_0^2 \frac{1}{x+4}$
2. $\int_0^{\ln 4} \exp(x) \cdot (\exp(x) - 2)^5$
3. $\int_{\pi/6}^{\pi/2} \frac{\cos x}{\left(\frac{1}{2} + \sin x\right)^4}$

Sample syllabus

Calculus I Course Information

Instructor:

Meeting Times and Location:

Office Hours:

Introduction:

Calculus, developed over 300 years ago, is an incredibly powerful tool with applications in a wide variety of disciplines from engineering to the social sciences. The key ideas are those of instantaneous rate of change, or derivative, and area under the curve, or integral. These ideas will be developed fully. In addition we will investigate the theoretical underpinnings of these ideas and some of their practical applications.

Prerequisites:

A student taking this course should have mastered basic algebra, including being able to simplify algebraic expression, solve equations, and operate with functions. The student should also have a basic knowledge of trigonometry. Typically, completion of a college level or high school level precalculus course or a combination of high school or college algebra and trigonometry should provide sufficient preparation.

Goals:

You will come to understand the basics of calculus. You will learn to solve problems and communicate your solutions clearly orally and in written form.

Topics covered:

1. Differentiation and applications
2. Antidifferentiation and applications
3. Integration and applications
4. The fundamental theorem of calculus
5. Limits

How the course will be conducted:

A common experience in many math classes is that the professor shows the students how to work certain problems. The solution is quite slick and is made to appear easy. The students are often given the impression that either; 1. there is no need for them to work any problems on their own since the solution methods have been so clearly demonstrated, or 2. anything short of the slick solution is of no value. Neither of these impressions is correct. Having watched someone else solve a problem does not guarantee that you will be able to solve similar problems. You learn how to solve problems only by actually working on them yourself. If you must watch someone else, it is far better that it be someone who is at your own level.

In order to become better problem solvers and also to help you better retain the essentials of calculus we will spend most of our class time working problems at the board and discussing these solutions.

In case you are a little suspicious about the legitimacy of all of this, I want to assure you that the style of teaching I am using has a long and distinguished tradition. My teaching style is influenced by what is often referred to as the Moore Method, Inquiry Based Learning, Discovery Based Learning, or the Socratic method. For more on this go to

<http://legacyrlmoore.org/method.html>

Problems will be posted at [insert electronic posting site]. You will not be given all the problems at once, but only enough to keep you busy for the next few class meetings. There are several reasons for this. First, it gives me a chance to tailor the problems to the class. Second, it gives you a chance to contribute problems by making your own conjectures and asking questions in class. And finally, it keeps me from spoiling your fun by giving away something that you might have discovered on your own.

Grades:

Board work and class participation 1/3

Written work 1/3

Final Exam 1/3

I also plan to give two mastery of basic skills tests. This is my way of checking whether you are ready for second semester calculus. I will insist that you get 70% or better on both of these tests before I give you a C or better in the class. I allow you three attempts to pass each test. The second and third attempt will have to be scheduled outside of class time. Let's all see if we can pass the first time.

Attendance:

I expect you to attend every single class meeting. The nature of the course requires you to be present, both to present your work to others and to respond to the work other students have presented (and hence to earn the points that constitute your grade). Of course, if urgent circumstances require your absence, you will be responsible for any work that you miss. In such cases, please get in touch with me as soon as possible.

More details on grades:

The board work and class participation grade will be based on a point system. You get 1 to 3 points for working a problem at the board and 1 to 2 points for contributing to someone else's solution. All points are counted in a positive sense. No one will ever get points taken away. You are never penalized for trying.

The point scale for board work is as follows:

0-You really have nothing. You are just winging it.

1-You have major errors.

2-You have some minor errors, your solution is not complete, or your presentation is not clear.

3-You have a clear, correct, and complete solution.

An unsuccessful first attempt with a score above 0 may be redeemed at a subsequent meeting. This will potentially allow you to get full credit for the problem in the end. While I will try to be as fair as possible in calling on students to present to the class and in awarding points for this work, I must ask you to respect my judgment on these matters.

You can also earn points by contributing in a significant way to class discussions.

Written work will primarily consist of careful write-ups of problems that you or someone else has presented in class. You will be expected to write thorough, correct, and clear solution to every problem that is worked at the board. Roughly once a week I will collect one or more of these written solutions from the previous week's in-class work to grade and return to you. You are to work on these write-ups alone. I will not accept written work on which you collaborated with someone else or that you copied from a book or the internet. These written solutions should contain considerably more detail and explanations than solutions presented at the board. Expect my standards to be quite high.

Written work is graded on a 0 to 5 scale, with point values as follows:

0-The work you handed in is not your own.

1-The work you handed in does not make sense or is irrelevant to the problem.

2-The work you handed in contains major errors that cannot be fixed without a complete overhaul.

3-Your idea is productive, but the details are incorrect or incomplete.

4-Your solution is essentially correct. What is keeping you from earning a 5 is either poor or imprecise wording or notation or minor computation or algebra errors.

5- Your solution is both correct and written in a clear and precise manner.

The grades are necessarily somewhat subjective, but I will do my best to make sure that you understand my assessment of your work.

Textbooks:

At the beginning of the semester I would prefer that you not consult any calculus textbooks, but I have no objection to you consulting algebra, trigonometry, or precalculus textbooks. For many this will be a relief and for others a burden. I want us talk and work with concepts and symbols that we all understand and to develop our own techniques for solving problem. Reading textbooks will, at least at the beginning, distract us from these aims.

Later on in the semester I see no harm in your using a textbook as a reference. Pretty much any calculus book will do, and it doesn't matter if it's old. You can probably pick one up for less than \$20.

Notes on the implementation of various versions of the problem sequence at Augusta State University 2002-2011

I began using an inquiry based problem sequence similar to this one to teach calculus starting with the fall semester of 2002. Classes ranged in size

from 9 to 29 students, majors coming from computer science, chemistry, physics, biology, mathematics, and a few others roughly in order of frequency from largest to smallest. Typically a few dual enrolled high school students took the classes as well. Grade distributions and success in follow up classes did not differ enough from other calculus classes taught at Augusta State University to draw any conclusions about the relative merits of using an IBL approach for this course. The actual data for the classes taught up through 2008 are as follows. My C or better rates in calculus I were 55% (60% 2005-2008) compared to departmental averages of 52% (56% 2005-2008). Students who took both calculus I and II with me had a 93% success rate (C or better) in a subsequent differential equations course compared to a departmental average success rate of 75%.

To the Student

This problem sequence is intended for a first semester course in calculus for STEM (science, technology, engineering, and mathematics) majors. The course is often designated as calculus I. The ideas on which calculus is based have been around for millennia, but it was in the seventeenth century that they finally coalesced into a form that could be practically applied and communicated to a wide audience. Since then these ideas have permeated most of the scientific disciplines and have contributed significantly to both theoretical research into the nature of our universe and to a host of very practical problem such as air travel and finding the right dose of a medicine.

This problem sequence is intended to guide you gently through the inner workings of calculus as well as to give you a glimpse of how calculus might be used to solve practical problems. It is my hope that you will come to understand not only what the tools and techniques of calculus are but that you will appreciate how and why they work. The best way to understand how something works is to build it yourself from scratch. As much as is possible given the time constraint, you will be guided through building calculus from the ground up. Having completed this task you will hopefully gain the confidence to use what you know without having to have an authority figure around to check your work and also to go on and tackle other areas of mathematics from the ground up.

The task you have ahead of you, worthwhile as it is, will not be easy. Real learning never is. You will find that often times you won't know how to get started. You may feel tempted to have someone hold your hand and show you exactly what to do. You may feel like you have been tossed in to the deep end of the pool without having had the benefit of swimming lessons. All of these are very natural feelings, especially since this might be the first time you have been given a mathematical task without someone first showing you a procedure for performing it.

What you may not realize is that you can actually create mathematics on your own and that you don't have to be a genius to do it. Once you discover this you will have gained so much in confidence that you may never need to have your hand held again. Once you know that you can figure out mathematics on your own there is nothing that can stop you from learning as much as you want. And when you learn this way you will realize that what

you created on your own you know in a much deeper way, in a way that you will not likely soon forget. You have made a giant step toward becoming an independent learner. This is ultimately what your professors want for you, because there will be a lot to learn when you leave the school you are now attending and venture out into the real world where you will not always have someone else to lean on.

You may find that you are spending more time and effort on this course than other students who are taking a similar course. You may feel that this is unfair. However, you may also find that the work you are doing for this course is more fun. And you might later discover you have an advantage in future courses that you take. You have learned to work independently and you have gained insight into the importance of knowing not just what to do but why it works and how it makes sense. I have had students tell me that they did not realize the value of what they were doing in my class until the end of the semester and sometimes even a long time after the semester was over. Just be patient. There are real rewards ahead. While the journey might be hard, it will also at times be extremely exhilarating. The first time you solve a difficult problem all on your own, you will know what I am talking about.

How then can you succeed at this new and difficult task of creating mathematics if you have never done it before? First you must read very carefully. Reading mathematics is not like reading fiction. Words in mathematics have very precise definitions. It is never good enough to say, “I think that means ...” It is really not important what you think a particular mathematical term means. The meaning is dictated by the definition. Now it might be that you don’t understand the definition. You will not be able to proceed until you do. Read the definition carefully. Are there words in the definition that you don’t understand? Then you will need to look these up. After you feel that you understand all the words in a definition you should see if you can come up with some examples that fit the definition. Maybe it will also be useful to come up with examples that do not fit the definition. Maybe you will need to draw some pictures. You will need to develop a considerable amount of patience. It might take hours or even days before you feel that you really understand a piece of written mathematics. All the while, you might be thinking that maybe you should just ask your instructor to help you understand what you are reading. This is certainly a reasonable last resort. A good instructor will help you figure out how to read the mathematics instead of outright explaining everything to you. That way, maybe next time you won’t have to rely on your instructor as much.

Besides becoming a better reader of mathematics you will also need to become a better problem solver. There are some things you can do to help you solve problems that you have never seen before. You can try to write down some examples. You can draw pictures. You can do some calcula-

tions. You can look for patterns. You can try to remember if you have ever seen something similar. Is there a simpler problem that is similar to the one you are trying to solve? If so, solve that one first. Read George Polya's *How to Solve it*. If you feel that you have made some progress, but you are not sure that you have a complete or correct solution, you should definitely take the opportunity to present what you have to the rest of the class. Many students are very reluctant to present solutions to problems that they are not sure they have solved correctly. This is understandable, but it is something you need to get over. There is no better learning experience than making a mistake, finding out why the mistake was made, and then using what you learned to produce a correct solution. This is not only a good experience for you, the presenter, but for the entire class that is the audience for your presentation. It is likely that many of your classmates made the same mistake or had the same false start. They too, want to know what went wrong, or why it was wrong. They will be learning along with you. Mistakes are some of your best teachers.

In this age of instant access to information on the internet, it is very tempting to search there for solutions or solution methods. There is plenty of information on the internet and much of it is quite useful. I learned how to repair my washing machine by watching a Youtube video. In this course you are not repairing a washing machine, you are trying to understand how and why a certain important body of mathematics works, so that you will understand it well enough to use it whenever you might need to. You are also trying to hone your skills as a problem solver and mathematical thinker. I am fairly sure that if I need to perform the same repair on my washing machine again I will need to watch the video again. Watching the video did not turn me into an appliance repair person. That would require having made many repairs to many different appliances, having made many mistakes, and understanding how many different appliances work. I want you to develop habits of mathematical thinking that will serve you well into the future, not to train you to be able to solve a limited number of very specific problems.

I think that the pride you get from working problems completely on your own will keep you from using the internet or other outside sources. And this way you will be able to reap the enormous benefits that this self-reliance can bring you.

Chapter 1

Motion and Lines

1.1 The Calculus of Motion

Often two quantities are related in a predictable way. For example if we roll a ball up a ramp then the distance, (measured in meters), from the ball to the starting point is related to the time (measured in seconds). In this case the quantity, distance, is related to the quantity, time. This particular relationship and many others have a mathematical name. Here distance is a **function** of time.

It is necessary to distinguish between the two quantities that are related by a function. One is typically called the independent variable, or input, and the other the dependent variable, or output. Some functions can be specified by a formula that tells us how the output is obtained from the input. If we give our function the name f then we might say something like: f is given by the formula $f(x) = x^2 + 2x$. Note that in this notation f is the function and $f(x)$ is the output value associated with the input value x . For the purposes of this course we will always choose our input values to be real numbers and our output values to be real numbers as well. Using this notation we can be a little more precise about what a function is: Suppose A is a set of real numbers. Then a function f with domain A is a set of ordered pairs of real numbers $(x, f(x))$ where each element from A appears once and only once as the first number in the pair. If we plot all the points $(x, f(x))$ belonging to a function in the plane we get what is called the graph of f .

The author performed an experiment in which a ball is rolled up an inclined ramp and some data for the function whose inputs are time (in seconds after the ball is released) and whose outputs are distance in meters from the starting point was collected.

Instructor note: The ball rolling experiment described here uses a simple wooden ramp made from a 1 by 8 piece of lumber 8 ft in length. Thin wooden strips (about 2 inches in width) are attached to the sides

of the board. These will keep the ball from rolling off of the sides of the ramp. The board is slightly inclined by propping up one end of the ramp slightly on some calculus textbooks (the only significant use I make of calculus textbooks in the class). A basketball is given a gentle push that sends it up the ramp just shy of the end, after which it comes rolling back down. I use a motion detector to record distance versus time data for the ball. Texas instruments makes the device that I use. It is called a “power ranger.” If you don’t have one, there is a good chance that your physics department does. While I, and at least some of my students get a kick out of actually running this experiment in class, you can also just use the data that is included in the problem sequence.

A table of some of the approximate time and distance values are given in the table that follows.

time (seconds)	distance (meters)
0	0
0.43	0.606
0.86	1.051
1.29	1.342
1.92	1.487
2.58	1.325
3.01	1.059
3.44	0.668
3.87	0.164
4	0

Problem 1. Find a formula of the form $f(x) = ax^2 + bx + c$ for a function f whose graph closely matches the ordered pairs given in the table. Sketch the graph of your function. (Hint: Start by plotting the points first.)

In what follows we will use a simplified model of a ball rolling experiment. We do this only so that the numbers don’t get in the way of the ideas that we are trying to understand. In this simplified model distance, (in meters), from the ball to the starting point is a function of the time (in seconds) that has elapsed since the ball was released. This function will have domain $[0, 4]$. (Recall that $[0, 4]$ means all numbers between 0 and 4 including 0 and 4 so that what we are saying here is that the ball rolling experiment takes place in 4 seconds and we start the clock at 0 seconds.) The function will be given by the formula $f(x) = -\frac{1}{2}x^2 + 2x$ where x is the time in seconds and $f(x)$ is the distance. Use this function in problems 2 to 10.

Instructor note: The instructor should make it clear to the students why a different position function is being used.

Problem 2. *At which time does the ball return to the point from which it was released?*

Problem 3. *At which time is the ball the farthest from the point from which it was released?*

Problem 4. *What is the total distance traveled by the ball?*

If we want to know how fast a moving object is traveling we can measure something called velocity. For an object traveling in a straight line the **average velocity** in some time interval is the change in position divided by the change in time. This quantity can be either positive or negative depending on the direction of motion and on how position is measured. In the rolling ball example our position is the distance to the point from which the ball was released. We used $f(x)$ for position and x for time.

Definition 5. *If f gives position as a function of time then the **average velocity** over the time interval from time a to time b is $\frac{f(b) - f(a)}{b - a}$.*

The units for velocity in this example are meters per second (m/s). In general the units for velocity are units of distance per units of time. Other examples include feet per second or miles per hour.

Problem 6. *Find the average velocity for the rolling ball function in the time interval from 1 to 2 seconds and in the time interval from 3 to 4 seconds.*

Problem 7. *Find a time interval in which the average velocity is 0 m/s.*

Instructor note: When asked to find a time interval in which the average velocity is 0 m/s students often observe that any time interval of the form $[2 - h, 2 + h]$ will work. If they make this observation I often include a problem in which they are asked to prove that this is in fact the case.

Problem 8. *Find a time interval in which the average velocity is 1 m/s.*

Problem 9. *Find a time interval in which the average velocity is -1 m/s.*

Problem 10. *Is there any time interval in the domain $[0, 4]$ for which the average velocity is 2 m/s? If so, give a time interval for which this is the case. If not, justify why no such time interval can exist.*

In problems 11 to 13 we will leave our rolling ball position function temporarily and consider more general position functions. For example you might imagine yourself walking in a straight line while a distance and time measuring device located on the same line as you measures distance to you and records time. The manner in which you walk will determine the shape of the distance versus time graph. Some ways of walking may match rather closely functions that have fairly simple formulas.

Problem 11. *Find the formula for a position function (position on a straight line as a function of time) which has the property that the average velocity is always 2 m/s no matter which time interval is chosen. Prove using definition 5 that the average velocity is always 2 m/s regardless of which interval is chosen.*

Problem 12. *Are there other position functions for which the average velocity is always 2 m/s? Give examples or prove that there are no others.*

Problem 13. *If a position function results in an average velocity that is always the same number no matter which time interval is chosen, what must be true about that position function? Justify your conclusion using definition 5.*

If we want to know how fast an object is traveling at some particular instant in time we run into a problem. Average velocity is computed by using two values of time. What does it mean to compute velocity using only one value of time? Suppose we want to find the velocity at time a . Trying $\frac{f(a) - f(a)}{a - a}$ doesn't make sense. What we could do instead is to consider some time b that is close to a . How close? Let's do some more problems. In problems 14 to 30 consider again the position function f that describes the position of a rolling ball and is given by the formula $f(x) = -\frac{1}{2}x^2 + 2x$ with domain $[0, 4]$.

Problem 14. *Find the average velocity in the time interval from 1 to 1.1.*

Problem 15. *Find the average velocity in the time interval from 0.9 to 1.*

Problem 16. *Find the average velocity in the time interval from 1 to 1.01.*

Problem 17. *Find the average velocity in the time interval from 0.99 to 1.*

Problem 18. *Experiment with some other small time intervals that start or end with 1. Comment on any pattern for the resulting average velocities.*

In problems 14 to 18 we considered time intervals starting with 1 and ending with some number very close to 1 and time intervals starting with a number very close to 1 and ending with 1. In problems 19 to 25 we will use the letter z to stand for our numbers that are close to 1. You should still consider z to be a number, just one that you would like to be allowed to be moved as close to 1 as you wish. For that reason we won't give it a specific value. As before we will be using the position function given by $f(x) = -\frac{1}{2}x^2 + 2x$ with domain $[0, 4]$.

Instructor note: You will notice that the next seven problems are very similar. After the first two have been done you can use this opportunity to call on students who have not yet presented anything in class.

Problem 19. Take some time that is close to 1 and call it z . Find the average velocity in the time interval from 1 to z in terms of z .

Problem 20. Reduce the expression obtained in problem 19 to lowest terms or use long division to simplify the expression.

Problem 21. Suppose z is some time that is close to 1 but smaller than 1. Find the average velocity in the time interval from z to 1 in terms of z and simplify the resulting expression.

Problem 22. Suppose z is some time that is close to 3 but larger than 3. Find the average velocity in the time interval from 3 to z in terms of z and simplify the resulting expression.

Problem 23. Suppose z is some time that is close to 3 but smaller than 3. Find the average velocity in the time interval from z to 3 in terms of z and simplify the resulting expression.

Problem 24. Suppose z is some time that is close to 2 but larger than 2. Find the average velocity in the time interval from 2 to z in terms of z and simplify the resulting expression.

Problem 25. Suppose z is some time that is close to 2 but smaller than 2. Find the average velocity in the time interval from z to 2 in terms of z and simplify the resulting expression.

Instructor note: This is a good place to stop when distributing the first installment of the problem sequence.

Definition 26. When we use the term *instantaneous velocity* at time a we are referring to a number that the average velocities in the time intervals $[z, a]$ and $[a, z]$ approach as z approaches a . We should be able to get as close to this number as we want by just making z close enough to a .

In the problems 27 to 30 use the expressions obtained in problems 19 to 25 to guess the instantaneous velocity of the ball at 1, 3, and 2 seconds.

Instructor note: Again there is considerable similarity in the next four problems and another opportunity presents itself to get reluctant students to the board.

Problem 27. Make a guess at the instantaneous velocity at time 1 second. Explain how you came up with your guess.

Problem 28. Make a guess at the instantaneous velocity at time 3 seconds. Explain how you came up with your guess.

Problem 29. Make a guess at the instantaneous velocity at time 2 seconds. Explain how you came up with your guess.

Problem 30. Suppose that a is some number between 0 and 4 make a guess at the instantaneous velocity at time a seconds. Explain how you came up with your guess.

Each of problems 14 to 30 dealt with the behavior of the average velocity in shrinking time intervals. One endpoint of the time interval remained fixed. The other endpoint approached the fixed endpoint. The problem lead us to examining the quotient $\frac{f(z) - f(a)}{z - a}$. A bit of algebraic manipulation often makes it possible to determine a single value that the quotient $\frac{f(z) - f(a)}{z - a}$ approaches as z approaches a .

Definition 31. The mathematical term given to the number that $\frac{f(z) - f(a)}{z - a}$ approaches as z approaches a is the **limit** of $\frac{f(z) - f(a)}{z - a}$ as z approaches a . The notation for this number is $\lim_{z \rightarrow a} \left(\frac{f(z) - f(a)}{z - a} \right)$.

A precise mathematical definition of $\lim_{z \rightarrow a} (\text{expression in } z)$ can be found in textbooks on calculus and in Appendix A of these notes. For now we will

use the rules given in appendix A to help us determine the limits that we need.

Instructor note: At this point the properties of limits in appendix A should be made available to the students and they should be encouraged to use them if they need them. If you have a particularly good class and time permits you may assign some of the problem from the appendix for extra credit. Otherwise they are there for the enrichment of interested students, particularly those that might be using this problem sequence outside of a classroom setting. For most students at this level it is perfectly fine to treat the limit rules as axioms. A formal definition of continuity does not play a significant role in this problem sequence. An instructor should feel free to use whatever intuitive concept of continuity they feel comfortable sharing with their students and direct students who desire more rigor to the material on continuity in the appendix.

Definition 32. Suppose that f is a function that gives the position of an object traveling in a straight line as a function of time then the *instantaneous velocity* at time a is the quantity $\lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a}$. The notation for this quantity is $f'(a)$. (Note that we could also say $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ or $f'(a) = \lim_{u \rightarrow a} \frac{f(u) - f(a)}{u - a}$. The particular name we give to the number that approaches a is irrelevant.)

Problem 33. Suppose f is the rolling ball function from before, given by the formula $f(x) = -\frac{1}{2}x^2 + 2x$ for $0 \leq x \leq 4$. Find the instantaneous velocity of the ball at the time 2.5 seconds into the roll.

Problem 34. Using the same position function from problem 33 find $f'(a)$ in terms of a .

Problem 34 shows us that given a position function f we can define a new function derived from f that gives us the instantaneous velocity at any time a in terms of a . Note that when we say $f'(a) = a + 2$ is a formula for the function f' then it is immaterial whether the letter for the variable is a, x, y, t , or anything else for that matter. The formula just tells us what to do with a given input. In other words, the function f' given by the formula $f'(a) = a + 2$ is the same function as the one given by the formula $f'(x) = x + 2$.

Problem 35. The formula for a position function is given. Find a formula for the corresponding instantaneous velocity function.

1. $f(x) = x^2$
2. $f(x) = 5x - 3x^2$
3. $f(x) = x^2 + x + 1$
4. $f(x) = 4x - 3$
5. $f(x) = \sqrt{x}$
6. $f(x) = x^3$ (Hint: $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$)
7. $f(x) = \frac{1}{x}$

Problem 36. Suppose that f is a position function and that the instantaneous velocity function v is given by $v(x) = 5$, in other words the instantaneous velocity is always 5 m/s. Find a possible formula for the function f . Show that for your function f the instantaneous velocity is indeed **always** 5 m/s.

Problem 37. Are there other position functions for which the instantaneous velocity is always 5? Either explain why there are none or give some examples.

As you saw in the ball rolling experiment, the instantaneous velocity can change. How rapidly it changes or whether it increases or decreases is something that you can determine by examining the instantaneous velocity function. If the velocity changes from time a to time b then the rate at which it changes and the direction, (positive or negative), of that change is given by the quantity $\frac{f'(b) - f'(a)}{b - a}$. Note that this is just the change in instantaneous velocity divided by the change in time. Another name for this quantity is the **average acceleration**. If the average acceleration in the time interval $[a, b]$ is positive then the instantaneous velocity is larger at b than at a . If the average acceleration in the time interval $[a, b]$ is negative then the instantaneous velocity is smaller at b than at a .

Problem 38. What are the units for average acceleration if the units for position are meters and the units for time are seconds?

Problem 39. Consider, once again, the ball rolling function f given by $f(x) = -\frac{1}{2}x^2 + 2x$. Find the average acceleration in three different time intervals contained in $[0, 4]$ of your choosing. What observations can you make about your results? See if you can prove that your observations are correct.

Problem 40. Consider the position function given by $f(x) = Ax^2 + Bx + C$ where A, B , and C are some constants and the domain is some interval. Find the average acceleration in any time interval of your choosing.

Problem 41. Suppose f is a position function which has a corresponding instantaneous velocity function given by $v(x) = 4x + 3$. What is a possible formula for f ? Check your formula by computing the instantaneous velocity function from your position function formula.

Problem 42. Suppose f is a position function which has a corresponding instantaneous velocity function given by $v(x) = mx + b$, where m and b are constants. What is a possible formula for f ? Check your formula by computing the instantaneous velocity function from your position function formula.

Problem 43. Instantaneous acceleration is the instantaneous version of average acceleration. Write down a mathematical definition of instantaneous acceleration at time a and use it to find the instantaneous acceleration at time 2 if the position function is given by $f(x) = x^3$.

Problem 44. Find the instantaneous acceleration at time a if the position is given by $f(x) = x^3$.

Problem 44 tells us how to obtain an acceleration function from a velocity function. The acceleration function is derived from the velocity function in much the same way that the velocity function is obtained from the position function. It is for this reason that if we start with the position function f and then derive the velocity function f' we call the acceleration function $(f')'$ or more simply f'' .

Problem 45. Find the instantaneous acceleration function for the ball rolling function we have been using (given by $f(x) = -\frac{1}{2}x^2 + 2x$).

Problem 46. Find the instantaneous acceleration function for the position function $f(x) = -x^2 + 3x + 5$.

Problem 47. Suppose a position function is given by $f(x) = Ax^2 + Bx + C$ where A, B , and C are some constants and the domain is some interval. Find the instantaneous acceleration function for this position function.

Problem 48. Suppose an object moving in a straight line experiences a constant instantaneous acceleration of 4m/s^2 . Assume time is measured in seconds and position in meters. Find a formula for a function that gives the position of this object. Are there other possible formulas? Explain.

Problem 49. Suppose an object moving in a straight line experiences a constant instantaneous acceleration of 4m/s^2 . Assume time is measured in seconds and position in meters. Suppose also that the position at time 0 is 1m and the instantaneous velocity at time 0 is 2 m/s. Find a formula for a function that gives the position of this object.

Problem 50. *A ball is rolled up a ramp. Let f be the function that gives the distance, (in meters), of the ball from the starting point as a function of time, (in seconds). Assume that the ball will experience a constant negative acceleration (deceleration) of -3m/s^2 . With what instantaneous velocity do we have to start the ball, (i.e. what must $f'(0)$ be), if we want the ball to roll 2m from the starting point and no further?*

Let us agree from this point forward that when we use the word velocity by itself we mean instantaneous velocity, and that when we use the word acceleration by itself we mean instantaneous acceleration.

It is a law of physics that an object traveling in a straight line will move at a constant velocity unless a force acts on it. It is another law of physics that a force produces an acceleration and that force and acceleration are related by the equation $F = ma$, where a is acceleration, m is mass, and F is force. It follows from this equation that if mass and force remain constant then acceleration will remain constant as well. In the absence of friction an object near the earth's surface experiences a constant gravitational force (in the direction perpendicular to the earth's surface) that produces a constant acceleration of -9.8 m/s^2 (the sign is negative because position is often measured as distance from some fixed ground level). Note that in the ball rolling experiment that we have been discussing the magnitude of the negative acceleration of the ball is significantly less than -9.8 m/s^2 since the ball is not moving in a vertical direction and hence the force that is accelerating the ball is the component of the gravitational force in the direction of the ramp. The laws of physics that I mentioned above were discovered by Sir Isaac Newton, who is also one of the discoverers of calculus. The best way to appreciate the profound beauty of the connections that Newton made is to make some of these connections yourself.

Problem 51. *Suppose a baseball is tossed straight up into the air from a height of 1.5 m and with an initial upward velocity of 30 m/s . Ignore friction and find the formula for a function that gives the height of the ball in terms of the number of seconds that have elapsed since the ball was released.*

Problem 52. *Consider the function obtained in problem 51. How high does the baseball go (i.e. what is its maximum height)?*

Problem 53. *If you are traveling in a car and you slam on the brakes to avoid hitting a possum then the distance your car travels before coming to a stop depends on two things; 1. the speed at which you were traveling when you slammed on the brakes, and 2. the amount of deceleration that your brakes produce, (Deceleration is just another word for negative acceleration. It means your velocity is decreasing.) Assume that your brakes produce a constant negative acceleration of -16 m/s^2 and that you are traveling 24*

m/s (about 54 mph) the instant you slam on the brakes. How far does your car travel before coming to a stop?

Problem 54. *Suppose you are driving in your car and you slam on your brakes at some point in time. Suppose also that your brakes cause your car to experience a constant deceleration from the time you slam on your brakes until you come to a complete stop. Find a formula that gives the distance your car travels from the time you apply the brakes until you come to a complete stop. This formula should be in terms of the following two quantities: 1. the speed you were traveling at the instant you applied the brakes and 2. the constant deceleration.*

Problem 55. *The width of a computer screen is 35.5 cm from left to right. A blue dot is moving in a horizontal line on the computer screen so that at time x seconds its distance from the left edge of the computer screen is $0.148x^3 - 3.4x^2 + 19.7x$ cm. Describe the motion of the blue dot in the time interval $[0, 15]$. Do this by indicating in which time intervals the dot is moving to the right and in which time intervals the dot is moving to the left.*

Instructor note: In the following problem students may need to be given a hint that they need to find the velocity function from the given position function.

Problem 56. *A plastic chicken moves back and forth along a straight track at the back of a pistol shooting booth at the county fair. Its motion is programmed by a computer so that the distance of the chicken from the far left end of the track at time x seconds after the start of a new shooting session is $2x^3 - 15x^2 + 36x + 5$ inches. Describe the motion of the chicken completely in the time interval $[0, 5]$. Do this by telling us in which time intervals the chicken is moving to the right and in which time intervals the chicken is moving to the left. Also indicate at which times the chicken changes direction.*

1.2 Tangent Lines

In problems 57 to 61 we will consider the graph of the function given by $f(x) = x^2$. Consider a line that goes through the point $(2, 4)$ on the graph of f . We will investigate what the slope of such a line can be if it intersects the graph of f at a point other than $(2, 4)$.

Instructor note: The next five problems should be completed before the definition of tangent line is revealed. That way investigating the

conditions on m that are necessary in order to ensure a unique intersection between the line $y = m(x - 2) + 4$ and the parabola $y = x^2$ becomes something more meaningful than a simple application of the definition of the tangent line.

Problem 57. Find a point on the graph of f other than $(2, 4)$. What is the slope of the line containing both your point and $(2, 4)$?

Problem 58. Consider an arbitrary point (x, x^2) on the graph of f other than $(2, 4)$. What is the slope of the line containing both (x, x^2) and $(2, 4)$ in terms of x ?

Problem 59. Let m be an arbitrary slope. Find the equation of a line through $(2, 4)$ with slope m . (Note that your equation will necessarily contain an m .)

Problem 60. Consider problem 59. Are there any values of m for which a line with slope m and containing $(2, 4)$ does not intersect the graph of f at a point other than $(2, 4)$? If so find such a value of m , if not explain why no such value of m can be found.

Problem 61. Is there a line through $(1, 1)$ that does not intersect the graph of f at any point other than $(1, 1)$?

Let us look at some pictures. Consider the function f given by the formula $f(x) = x^2$. Suppose $f(x)$ gives position at time x . We want to get a picture of $f'(1)$. Recall that $f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$. The quantity $\frac{f(x) - f(1)}{x - 1}$ represents the slope of the line through the points $(1, f(1))$ and $(x, f(x))$. As x approaches 1 the line through the points $(1, f(1))$ and $(x, f(x))$ starts to look more and more like the line that touches the graph of f at only the point $(1, 1)$. This line is called the tangent line to the graph of f at the point $(1, 1)$. The slope of this tangent line is $f'(1)$. In general if f is any function, the slope of the tangent line to the graph of f at the point $(a, f(a))$ is $f'(a)$. (Note that this process only works if $f'(a)$ is defined.)

Instructor note: Some kind of visual display (e.g. an animation) illustrating secant lines approaching the tangent line can be quite helpful at this point.

Problem 62. Let $f(x) = \sqrt{x}$. Find the equation of the tangent line to the graph of f at the point $(4, 2)$. Sketch the graph of f and the tangent line on the same coordinate system.

Problem 63. Let $f(x) = x^3$. Find the equation of the tangent line to the graph of f at the point $(1, 1)$. Sketch the graph of f and the tangent line on the same coordinate system.

Problem 64. Let $f(x) = \frac{1}{x}$. Find the equation of the tangent line to the graph of f at the point $(2, \frac{1}{2})$. Sketch the graph of f and the tangent line on the same coordinate system.

Problem 65. Find the formula for a quadratic function whose graph describes the cross-section of a curved roof with slope 1 at the bottom and slope $\frac{1}{6}$ at the top and where the straight-line distance between bottom and top is 20 feet. In mathematical language what I am looking for is a quadratic function for which the slope of the tangent line to the graph is 1 at one point and $\frac{1}{6}$ at another point and those two points are 20 units apart. (Hint: Use $(0, 0)$ for the point where the slope of the tangent line is 1. See Figure 1.1 below.)

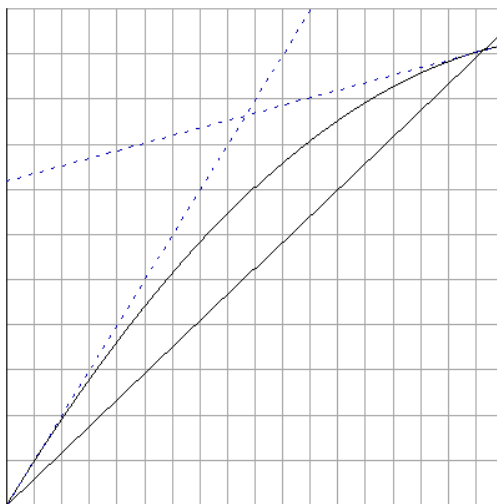
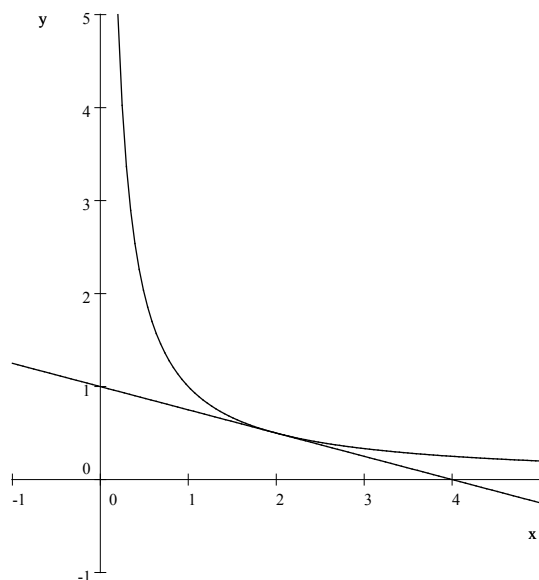


Figure 1.1: Roof

Problem 66. Is there an equation of a line that goes through the point $(4, 11)$ and is tangent to the graph of the function given by $f(x) = x^2$ at some other point? If so, find that equation. If not, explain why not.

Problem 67. For which points (x, y) in the coordinate plane is it possible to find a line that is tangent to the graph of the function given by $f(x) = x^2$ and that also goes through (x, y) ? Justify your answer.

Figure 1.2: Tangent line to $y = 1/x$ graph

Problem 68. Suppose a is some positive number. The line tangent to the graph of the function given by $f(x) = \frac{1}{x}$ at the point $\left(a, \frac{1}{a}\right)$, the x -axis, and the y -axis form a right triangle (see Figure 1.2 showing the case where $a = 2$). Find the area of this triangle for three different positive values of a . Comment on and justify any pattern that you observe.

The tangent line to the graph of f at the point $(a, f(a))$ can be used to make approximations. It is the one line that best approximates or is closest to the graph of f for x values close to a . Since lines are much simpler and easier to deal with computationally, we often use the tangent line instead of the function to make estimates.

Instructor note: The next three problems ask students to use the tangent line to approximate values of a function close to the point of tangency. These problems should be preceded with a classroom demonstration showing a graph of a function and the graph of its tangent line, followed by zooming in on the point of tangency to bring home the message that the two graphs are nearly indistinguishable near the point of tangency. It is also worth pointing out the advantage of using a linear approximation in those cases where machine computations fail to give satisfactory results.

Problem 69. Let $f(x) = \sqrt{x}$. Find the tangent line to the graph of f at the point $(100, 10)$. Use the tangent line to get a quick estimate of $\sqrt{100.1}$. Just

for fun, compare your estimate with a calculator estimate. How close are the two estimates to each other?

Problem 70. Let $f(x) = \frac{1}{x}$. Find the equation of the tangent line to the graph of f at the point $(1, 1)$ and use it get an estimate of $\frac{1}{1.00000001}$.

Problem 71. The volume of a solid aluminum cube with side length 2 cm is 8 cm^3 . Now suppose that a slight increase in temperature causes the cube to expand so that each side becomes 10^{-20} cm longer. Approximate the increase in the volume using the tangent line to the graph of the function given by $f(x) = x^3$.

Chapter 2

Derivative Formulas and Applications

2.1 Elementary Formulas for Taking Derivatives

Let us introduce some terminology. We have used the quantity $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ to find instantaneous velocity and to find the slope of the tangent line. This quantity has many other applications but the term we use for it no matter where it might turn up is the **derivative of f at a** . The function that takes the input a and yields the output $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ is called the derivative function or simply the **derivative**. The notation for the derivative of f is f' . If g is the name of the function then the name of the derivative function is g' . If f' is the function then the derivative of f' is $(f')'$. We call this last function the **second derivative** and use the shorthand f'' . Similarly we call $(f'')' = f'''$ and so on until we get tired of all those quotation marks and just write $f^{(n)}$ for the n th derivative.

The process of starting with a function and then finding the derivative function is called differentiation. Unless you have already taken calculus, what you know so far is that this process involves taking the quotient $\frac{f(x) - f(a)}{x - a}$, massaging it with algebra until it takes on a nicer form that is the same everywhere except maybe at $x = a$, and then evaluating the limit as x approaches a .

If you do this process enough times to enough different functions then some patterns begin to emerge. The patterns get us out of having to do all that algebra and take all those limits every time we need a derivative. I will let you discover some of the patterns and tell you about others. In the end you will have enough tools to differentiate just about any function for which a formula involving arithmetic, and known elementary (e.g. trigonometric, exponential, etc.) functions exists.

In problems 72 through 78 the formula for a function f will be given.

You are to find a formula for f' and prove that your formula works using only the definition of the derivative as a limit of a quotient and the properties of limits that you have been given.

Problem 72. $f(x) = c$.

Problem 73. $f(x) = x$.

Problem 74. $f(x) = x^n$ where n is any positive integer.

Problem 75. $f(x) = \sqrt{x}, x > 0$.

Problem 76. $f(x) = \frac{1}{x}, x \neq 0$.

Problem 77. $f(x) = cg(x)$ where c is a constant and g is a function. Give your formula for $f'(x)$ in terms of c and $g'(x)$.

Problem 78. $f(x) = g(x) + h(x)$. Give your formula for $f'(x)$ in terms of $g'(x)$ and $h'(x)$.

Problem 79. In each of the following a formula for f is given. Use the rules developed in problems 72 through 78 to find a formulas for f' , f'' , and f''' .

1. $f(x) = 5x^3 + 2x^2 - 11$

2. $f(x) = 12\sqrt{x} + \frac{4}{x}$

3. $f(x) = (x+2)(x-3)$

4. $f(x) = \frac{x}{3} - \frac{x^3}{5} + 1$

5. $f(x) = ax^4 + bx^3 + cx^2 + dx + e$

The next few formulas for derivatives are a little bit harder to derive. You will be asked to prove them under certain special conditions and then we will assume they work in all other cases in which all the expressions involved are defined.

Problem 80. Assume that $\lim_{x \rightarrow a} g(x) = g(a)$, and $\lim_{x \rightarrow a} h(x) = h(a)$. Let $f(x) = g(x)h(x)$. Show that $f'(a) = g(a)h'(a) + g'(a)h(a)$. (Hint: add and subtract $g(a)h(x)$ in the numerator of the expression $\frac{g(x)h(x) - g(a)h(a)}{x - a}$.) This is called the **Product Rule**.

Problem 81. Assume that $\lim_{x \rightarrow a} g(x) = g(a)$, and $\lim_{x \rightarrow a} h(x) = h(a)$ and that $h(x)$ is never equal to 0. Let $f(x) = \frac{g(x)}{h(x)}$ and show that

$$f'(a) = \frac{g'(a)h(a) - h'(a)g(a)}{(h(a))^2}.$$

This is called the **Quotient Rule**.

Problem 82. Assume that $\lim_{x \rightarrow a} h(x) = h(a)$, that $\lim_{x \rightarrow a} g(h(x)) = g(h(a))$, and that $\lim_{x \rightarrow a} \frac{g(h(x)) - g(h(a))}{h(x) - h(a)} = g'(h(a))$. Show that $f'(a) = g'(h(a)) \cdot h'(a)$. This is called the **Chain Rule**.

Problem 83. In each of the following a formula for f is given. Use the rules given immediately above and those developed in previous problems to find a formula for f' .

1. $f(x) = (2x - \frac{1}{x} + 2)(x^3 - x^2 + 4x + 3\sqrt{x})$ (Here I want you to use the Product Rule and not expand the product first. You will not be required to simplify your derivative.)

2. $f(x) = \frac{1}{1+x^2}$

3. $f(x) = \frac{x}{1+x^2}$

4. $f(x) = \frac{ax+b}{cx+d}$, (Here a, b, c , and d are constants.)

5. $f(x) = \sqrt{x^2+1}$

6. $f(x) = (x^3+9)^6$

7. $f(x) = \frac{x}{\sqrt{1+3x^2}}$

8. $f(x) = \sqrt{1 + \sqrt{1 + \sqrt{x}}}$

Problem 84. Find a formula for f'' for each f given in the previous problem.

2.2 Graph Sketching

Instructor note: Before giving the students the problems in which they use the sign of the derivative to get slope information, I recommend a brief review of solving polynomial and rational inequalities.

The sign (plus or minus) of the derivative can help us sketch the graph of f . If the derivative is always positive in some interval then the tangent line has positive slope and the graph of f will rise from left to right. Figure 2.1 shows the graph of the function given by $f(x) = 3x - x^3$ with tangent lines at $x = -2$, $x = -0.5$, and $x = 2$.

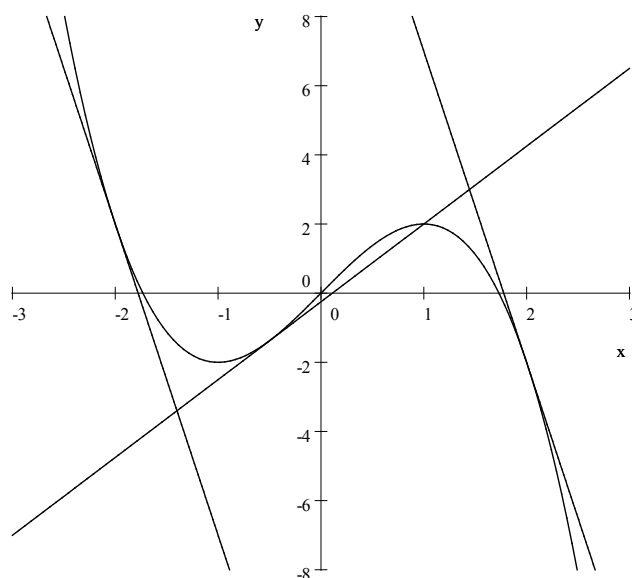


Figure 2.1: Tangents to the graph of $y = 3x - x^3$

Problem 85. Let $f(x) = 3x - x^3$, show that $f'(x)$ is positive for $-1 < x < 1$ and negative for $x > 1$ or $x < -1$.

Problem 86. Let $f(x) = 20x^3 - 63x^2 + 66x + 12$. Find out for which values of x , $f'(x)$ is positive and for which values of x , $f'(x)$ is negative. Use this information to sketch a graph of f that shows the important features of the graph, (e.g. places where the graph changes direction).

Problem 87. Let $f(x) = x^2 + \frac{1}{x}$. Find out for which values of x , $f'(x)$ is positive and for which values of x , $f'(x)$ is negative. Use this information to sketch a graph of f that shows the important features of the graph, (e.g. places where the graph changes direction).

Problem 88. Use the rules that you already know for taking derivatives to prove the following rule: If $f(x) = x^n$ and n is a negative integer then $f'(x) = nx^{n-1}$. (Note that we already know that this works if n is a positive integer. To avoid confusion you might want to let $n = -k$ where k is a positive integer.)

2.3 Derivatives of Trigonometric Functions

Instructor note: A brief review of trigonometry, particularly the definitions of the trigonometric functions, is in order right before the problems that involve limits and derivatives of trig functions.

Problems 90 through 106 will help us get derivative rules for the trigonometric functions.

Let us first say a little more about limits. When we say $\lim_{x \rightarrow a} f(x) = L$ we mean that we can make $f(x)$ as close to L as we want as long as we take x to be close enough to but not necessarily equal to a . It does not matter whether x happens to be larger or smaller than a . In either case $f(x)$ can be made close to L . If we want to specify that we only want to consider x values larger than a , we use the terminology, “the limit as x approaches a from the right” and the notation $\lim_{x \rightarrow a^+} f(x)$. Similarly if we only want to consider x values smaller than a , we use the terminology, “the limit as x approaches a from the left” and the notation $\lim_{x \rightarrow a^-} f(x)$. It is a fact that $\lim_{x \rightarrow a} f(x) = L$ if and only if both $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$ and sometimes the only way to get at $\lim_{x \rightarrow a} f(x)$ is to look at $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ separately.

In what follows we will need the following theorem which is also stated under Properties of Limits in Appendix A.

Instructor note: A graphical illustration of the Squeeze Theorem is recommended at this point

Theorem 89 (The Squeeze Theorem). Suppose that f, g , and h are all defined on the same open interval (c, d) containing a except possibly at a and that for all values of x in (c, d) except possibly at a the inequality $f(x) \leq g(x) \leq h(x)$ holds. If a is in (c, d) and both $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} h(x) = L$ then it must also be true that $\lim_{x \rightarrow a} g(x) = L$. A similar statement applies to left and right hand limits.

Problem 90. The Figure 2.2 shows a circle with radius 1 and center B , and some line segments. The angles ACB and DEB are right angles. Let θ be the measure of angle ABC in radians.

1. Show that the area of the sector enclosed by segments $\overline{BD}$ and $\overline{BC}$ and arc $\widehat{CD}$ is $\frac{\theta}{2}$.
2. Show that the area of triangle ABC is $\frac{\tan \theta}{2}$.
3. Show that the area of triangle BDC is $\frac{\sin \theta}{2}$.

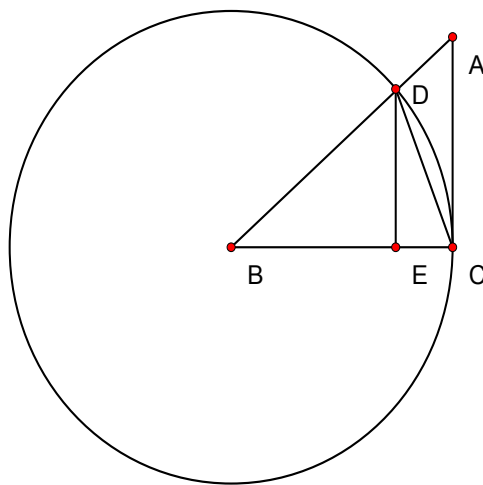


Figure 2.2: Unit Circle

Problem 91. Explain why Figure 2.2 gives you the inequality $\frac{\sin \theta}{2} \leq \frac{\theta}{2} \leq \frac{\tan \theta}{2}$ for all $0 < \theta < \frac{\pi}{2}$.

Problem 92. Note that if $-\frac{\pi}{2} < \theta < 0$ then $0 < -\theta < \frac{\pi}{2}$. Use this and the fact that $\sin$ and $\tan$ are odd functions to show that

$$\frac{\tan \theta}{2} \leq \frac{\theta}{2} \leq \frac{\sin \theta}{2}$$

for $-\frac{\pi}{2} < \theta < 0$.

Problem 93. Show that $\lim_{\theta \rightarrow 0^+} \sin \theta = 0$. (Hint: Use the right handed version of the Squeeze Theorem and the fact that $\sin \theta \geq 0$ for $0 \leq \theta \leq \frac{\pi}{2}$.)

Problem 94. Show that $\lim_{\theta \rightarrow 0^-} \sin \theta = 0$. (Hint: Use the left handed version of the Squeeze Theorem and the fact that $\sin \theta \leq 0$ for $-\frac{\pi}{2} \leq \theta \leq 0$.)

Problems 93 and 94 together allow us to conclude that $\lim_{\theta \rightarrow 0} \sin \theta = 0$.

Problem 95. Use the identity $(\cos \theta)^2 = 1 - (\sin \theta)^2$ and the fact that $\cos \theta \geq 0$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ to show that $\lim_{\theta \rightarrow 0} \cos \theta = 1$.

Problem 96. Derive the inequality $\cos \theta \leq \frac{\sin \theta}{\theta} \leq 1$ for all $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ except $\theta = 0$ using the inequalities from problems 91 and 92. (Hint: Recall that $\tan \theta = \frac{\sin \theta}{\cos \theta}$)

Problem 97. Use problems 95 and 96 and the Squeeze Theorem to show that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$.

Problem 98. Verify the identity $\frac{\cos \theta - 1}{\theta} = \frac{\sin \theta}{\theta} \cdot \frac{(-\sin \theta)}{(\cos \theta + 1)}$.

Problem 99. Use problems 97 and 98 and the fact that the limit of a product is the product of the limits (see Properties of Limits which is a separate part of these notes) to show that $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$.

Problem 100. Let $f(x) = \sin x$. Then $f'(0) = \lim_{x \rightarrow 0} \frac{\sin x - \sin 0}{x - 0}$. What is this limit?

If we want to find the derivative of the sine function at some number a other than 0 then we must compute the following limit:

$$\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a}$$

Since we know limits of the sine and cosine function at 0 we conveniently move the graph over so that the point $(a, \sin a)$ is moved to $(0, \sin a)$. Figure 2.3 shows the situation when $a = 2$.

The solid curve is the sine graph and the dotted curve is the sine graph after a left shift by 2. Notice that the slope of the tangent line at 0 for the shifted graph is the same as the slope of the tangent line at 2

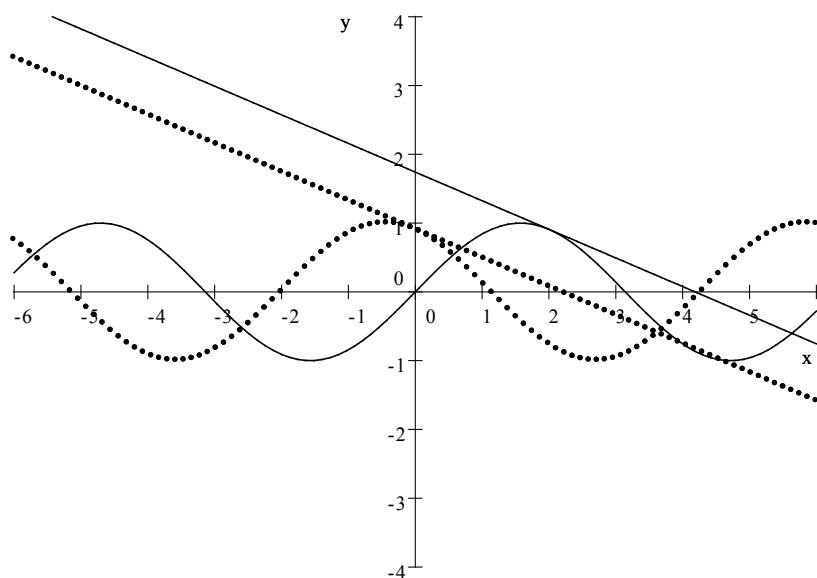


Figure 2.3: Shifted sine graph

for the original graph. Note also that shifting the graph so that the point $(a, \sin a)$ is moved to $(0, \sin a)$ amounts to graphing the function g given by the formula $g(x) = \sin(x + a)$. This illustrates the fact that:

$$\lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a} = \lim_{x \rightarrow 0} \frac{\sin(x + a) - \sin(0 + a)}{x - 0} = \lim_{x \rightarrow 0} \frac{\sin(x + a) - \sin(a)}{x}$$

The third limit we can compute with the tools that we have developed.

Problem 101. Recall that $\sin(x + a) = \sin x \cos a + \cos x \sin a$. Use this identity and the limits we just developed to justify the following:

$$\lim_{x \rightarrow 0} \frac{\sin(x + a) - \sin(a)}{x} = \left[\lim_{x \rightarrow 0} \frac{\sin x}{x} \cos a + \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} \sin a \right] = \cos a$$

Problem 101 gives us the following derivative rule:

$$\text{If } f(x) = \sin x \text{ then } f'(x) = \cos x.$$

Problem 102. Use the identity $\cos x = \sin\left(\frac{\pi}{2} - x\right)$ and the chain rule to show that if $f(x) = \cos x$ then $f'(x) = -\cos\left(\frac{\pi}{2} - x\right)$ which is the same as $-\sin x$ by another identity.

Problem 103. Show that if $f(x) = \tan x$ then $f'(x) = (\sec x)^2$. (Hint: Use the identity $\tan x = \frac{\sin x}{\cos x}$.)

Problem 104. Show that if $f(x) = \sec x$ then $f'(x) = \sec x \tan x$.

Problem 105. If $f(x) = \cot x$ what is $f'(x)$? Prove your result.

Problem 106. If $f(x) = \csc x$ what is $f'(x)$? Prove your result.

2.4 Derivatives for Negative and Rational Powers

Recall that an identity is an equation that is true for all values of the variable for which both sides make sense. For example $(\sin x)^2 = 1 - (\cos x)^2$ is an identity. So is $(\sqrt{x})^2 = x$.

We can use identities in calculus to develop derivative rules. Note that identities have the form $f(x) = g(x)$ where f and g are functions. The identity tells us that for all values of x in the domain of both f and g the two functions have the same value. In other words if the domains of f and g are the same we can say that f and g are in fact the same function. Certainly if two functions are the same then they will have the same derivative. It follows that if $f(x) = g(x)$ is an identity then so is $f'(x) = g'(x)$. We can use this fact to obtain derivative rules. The trick is to "differentiate both sides" of an identity and then solve for an unknown derivative. For example suppose we want a derivative rule for the function given by $f(x) = x^{1/5}$. For this function $(f(x))^5 = x$ is an identity. If we differentiate both sides and use the chain rule we get $5(f(x))^4 \cdot f'(x) = 1$. We don't know a formula for $f'(x)$ but we know this formula must satisfy the identity $5(f(x))^4 \cdot f'(x) = 1$. If $f(x)$ is not 0 this is equivalent to the identity $f'(x) = \frac{1}{5(f(x))^4} = \frac{1}{5(x^{1/5})^4} = \frac{1}{5}x^{-4/5}$. Note that $f(x) = 0$ precisely when $x = 0$. Thus we get $f'(x) = \frac{1}{5}x^{-4/5}$ for $x \neq 0$.

Problem 107. We have shown that if $f(x) = x^n$ and n is an integer then $f'(x) = nx^{n-1}$. Now we will show that this rule works also for exponents of the form $\frac{1}{n}$ where n is a non-zero integer. Let n be a non-zero integer and suppose $x > 0$ if n is even. If $f(x) = x^{1/n}$ show that $f'(x) = \frac{1}{n}x^{(1/n)-1}$. (Hint: Differentiate both sides of an identity, see discussion above.)

Problem 108. Now we show the power rule works for any rational number. A rational number is any number that is equal $\frac{a}{b}$ for some integers a and b with $b \neq 0$. Suppose that r is a rational number and $f(x) = x^r$. Prove that $f'(x) = rx^{r-1}$. (Hint: Let $x^r = x^{\frac{a}{b}} = \left(x^{\frac{1}{b}}\right)^a$ and use the chain rule.)

2.5 Harmonic Motion

Instructor note: The problems involving the motion of an object attached to a spring provide another opportunity to use a motion detector and a mass spring system to collect some real data and compare it to the theory or to use the theory to compute spring constants or masses from the period. When the students are working the harmonic motion problems they should be reminded of the problems involving motion under the assumption of constant acceleration. The idea of starting with a reasonable model function and solving for unknown coefficients is a recurring theme.

Next we revisit our study of motion. Suppose an object is suspended from a spring. Hooke's Law tells us that the force acting on the object is proportional to the displacement of the object from its equilibrium position (where it hangs when it is not disturbed). Now suppose we set an object suspended from a spring in motion by pulling it down and then releasing it. If f is the function that gives the displacement of the object as function of time (and other forces such as friction are ignored) then at any time x the force acting on the object is some negative constant times $f(x)$ (i.e. $-kf(x)$ where $k > 0$). The magnitude of the constant k depends on the particular spring and is called the spring constant. Since Newton's second law of motion tells us that force is equal to mass times acceleration, and acceleration at time x is $f''(x)$, we get $mf''(x) = -kf(x)$ where m is the mass of the object.

Problem 109. *It turns out that if $f(x) = A \sin(B(x - C))$ and B is chosen in the right way then we can indeed get $mf''(x) = -kf(x)$. Find B in terms of m and k so that $mf''(x) = -kf(x)$. (Remember to use the chain rule.)*

Problem 110. *Suppose a pumpkin with mass 5 kg is attached to a spring with spring constant k equal to 20 N/m (N here stands for Newtons which is a unit of force.) Suppose we start this pumpkin in motion by pulling it to 0.3 m below its equilibrium position and letting it go without giving it a push. Find numbers A , B , and C so that $f(x) = A \sin(B(x - C))$ is the displacement of the pumpkin from equilibrium at time x seconds after it is released. Ignore friction. (Hint: $f(0) = -0.3$ and $f'(0) = 0$)*

Problem 111. *Take the same pumpkin attached to the same spring as in problem 110. Now start it from equilibrium position with a push upward that gives it an initial velocity of 1.2 m/s. Find the displacement in terms of time by using a formula of the form $f(x) = A \sin(B(x - C))$.*

Problem 112. *Take the pumpkin one last time and start it from 0.8 m below equilibrium with an upward initial velocity of 1.6 m/s. Find a correct formula of the form $f(x) = A \sin(B(x - C))$ for displacement at time x .*

The numbers A , B , and C in the formula $f(x) = A \sin(B(x - C))$ determine the amplitude, period, and phase shift respectively of the graph of f . The number $|A|$ is the amplitude, the number $\frac{2\pi}{B}$ is the period, and the number C is the phase shift. The phase shift represents a horizontal shift of the standard sine graph. If $C > 0$ then the graph of f is a right shift of the standard sine graph. If $C < 0$ then the graph of f is a left shift of the standard sine graph.

Problem 113. Assume $f(x) = A \sin(B(x - C))$ is displacement from equilibrium as a function of time x for an object suspended from a spring. Determine how each of the three quantities amplitude, period, and phase shift depend on the mass, spring constant, initial velocity $f'(0)$, and initial displacement $f(0)$.

Problem 114. Find the equation of the tangent line to the graph of the sine function at the point $(0,0)$ and use it to approximate $\sin(.00123)$. How close is your approximation to the value your calculator gives you for $\sin(.00123)$?

2.6 More Graph Sketching

The first and second derivatives of a function f can be used to sketch the graph of f . The sign of f' tells us about the slope. Sign changes in f' indicate possible extreme points. Sign changes in f' typically occur at the zeroes of f' , or at values of x where the derivative is not defined. The sign of f'' tells us whether the slope is increasing or decreasing. If the slope is increasing the graph is said to be concave up. If it is decreasing the graph is said to be concave down. Sign changes of f'' indicate changes in concavity. A point on the graph of f at which the concavity changes is called an **inflection point** of f . See if you can locate the extreme point(s) and the inflection point(s) in Figure 2.4.

Problem 115. Consider the function f given by $f(x) = \frac{1}{1+x^2}$. Find any and all extreme points and inflection points of the graph of f . Determine for which values of x the slope of the graph is positive and for which values of x the slope is negative. Finally, determine for which values of x the graph of f is concave up and for which values of x the graph is concave down. Use this information to sketch a graph of f . Compare your sketch with a machine (calculator or computer) generated graph.

Problem 116. Sketch graphs of the functions whose formulas are given below using only first and second derivative information. Your graph should

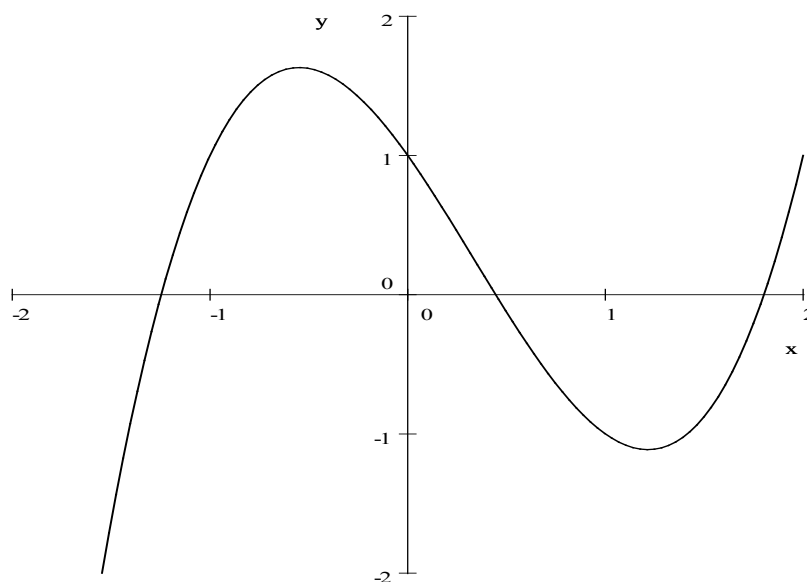


Figure 2.4: Point of inflection

show all local extreme points and all inflection points, or in the case of the functions whose formulas involve trigonometric functions, enough of them to see what the pattern is.

1. $f(x) = x^4 - 2x^2 + 3$
2. $f(x) = x^3 - 12x + 1$
3. $f(x) = x^{2/3}(6 - x)^{1/3}$ (This one requires some patience. Work carefully, remember to use the product rule when necessary, and simplify before going on to take the second derivative.)
4. $f(x) = \frac{x^2}{x^2 + 4}$
5. $f(x) = \sin x + \frac{x}{2}$
6. $f(x) = \sin x + x$
7. $f(x) = \sin x + 2x$
8. $f(x) = x - \frac{x^3}{6} - \sin x$
9. $f(x) = 2 \tan x - (\sec x)^2$

2.7 Optimization

One practical application of calculus is in finding the optimal value of some quantity that might be important to us, (e.g. cost, profit, gas mileage, distance, strength, truth, beauty, happiness). In many cases the optimal value turns out to be the largest or smallest value of some function. Finding this value or the associated best input is called **optimization**. You may have solved optimization problems before in an algebra or precalculus class. In that setting the problem amounted to finding the right function and then either graphing it on a calculator, or in the special case that your function was quadratic, finding the vertex of the graph of the function. Calculus gives us another tool for finding maximum or minimum values of functions. We look for what are called local **extreme points**. This amounts to looking for sign changes in the derivative. The beauty of the calculus technique is that you can find extreme points in terms of unspecified constants. A machine generated graph is not possible in these cases so we must rely on calculus instead.

Problem 117. Find a number x for which the sum of the squares of the distances between x and each of the numbers 1, 2, 12, 15, and 20 is as small as possible.

Problem 118. Recall that the mean $\bar{x}$ of n numbers $x_1, x_2, \dots, x_n$ is given by $\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$. Show that $\bar{x}$ minimizes the sum of the squares of the distances between $\bar{x}$ and each of the numbers $x_1, x_2, \dots, x_n$.

Problem 119. What quantity is minimized by the median?

Problem 120. A club is trying to sell T-shirts on campus. Research has shown that the relationship between the number N of T-shirts sold and the price x is given approximately by $N = 500 - 20x$. If the cost of N shirts is $100 + 4N$ dollars, what price yields the maximum profit?

Problem 121. Repeat problem 120 with the new assumption that $N = \frac{2000}{x} - 20$.

Problem 122. A truck driver gets paid \$1.00 per mile but must pay her own fuel expenses. Her fuel economy is a function of her speed. Suppose that she gets 8 miles per gallon at 40 mph and that her fuel economy goes down $\frac{1}{10}$ of a mile per gallon for every 1 mile per hour increase in her speed. Assuming that the trucker pays \$4.00 per gallon of diesel fuel, at what speed should the truck driver travel in order to get the best net pay per hour (hourly pay minus hourly fuel expenses). What is the maximum net pay per hour that the truck driver receives?

Problem 123. Refer to problem 122. Suppose that the cost of fuel goes up to \$5.00 and the pay per mile does not go up, what is now the maximum net pay that the truck driver can receive?

Problem 124. Refer to problems 122 and 123. Suppose that the trucking company would like to make sure that the truck driver's maximum net pay remains the same regardless of what the cost of fuel is. Determine the pay per mile in terms of the price per gallon so that the maximum net pay stays the same.

Problem 125. A cardboard box is constructed by taking a rectangular piece of cardboard with dimensions w and l , removing four squares of the same size from each of the four corners and then folding up the four smaller rectangles that lie between the squares. See Figure 2.5. The folding takes place along the dashed lines. Find the length of the squares that must be cut out (in terms of w and l) in order for the resulting box to have the largest possible volume.



Figure 2.5: Layout for folding

Problem 126. What is the relationship between the height and the radius of a cylindrical can that has the smallest surface area, (including top and bottom), for a given volume? Justify your answer.

Problem 127. Find the point on the parabola $y = x^2$ that is closest to the point $(2, 3)$. Let us call the point that you found (a, a^2) . Show that the line through the two points (a, a^2) and $(2, 3)$ is perpendicular to the line tangent to the graph of $y = x^2$ at the point (a, a^2) .

Problem 128. If a rubber ball is dropped from a certain height x then the height that it reaches after the first bounce is more or less directly proportional to the height from which it was dropped. In other words, if y is the height after the first bounce then $y = kx$ where k is some constant. Record some data (some pairs of numbers (x, y)) for a particular rubber ball. Find a number k so that the line $y = kx$ best fits the data in the sense that the sum of the squares of the vertical distances between the data points and the line is minimized.

Problem 129. Suppose that $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ are data points collected for a relationship that is approximately of the form $y = kx$. Find the constant k in terms of the numbers $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ so that the sum of the squares of the vertical distances between the points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ and the line $y = kx$ is minimized.

2.8 Derivatives of Inverse Trigonometric Functions

Just a few more derivative rules before we proceed to another topic. The arcsin function is the function with the property that $\arcsin(\sin(x)) = x$ for all x in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. The domain of the arcsin function is $[-1, 1]$, as it must be in order to make sense of this definition. You can easily check that $\arcsin(-1) = -\frac{\pi}{2}$, $\arcsin(1) = \frac{\pi}{2}$, $\arcsin(0) = 0$, and $\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$. You can also check that for any number x in $[-1, 1]$, $\sin(\arcsin(x)) = x$. It is this last fact that we use to develop a derivative rule for the arcsin function.

Problem 130. Show that $\cos(\arcsin x) = \sqrt{1-x^2}$. (Hint: In Figure 2.6 $h(x) = \arcsin(x)$.)

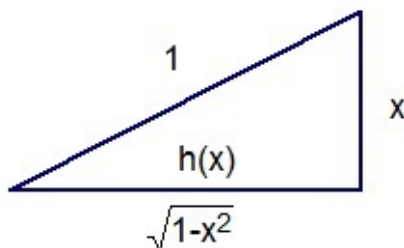


Figure 2.6: Right Triangle

Problem 131. Letting $h(x) = \arcsin(x)$ to simplify notation, and noting from the discussion above that $\sin(h(x)) = x$ is an identity for all x in $[-1, 1]$, find a formula for $h'(x)$.

Problem 132. Derive a formula for the derivative of the arctan function.

Problem 133. Derive a formula for the derivative of the arccos function.

Problem 134. Derive a formula for the derivative of the arcsec function.

Chapter 3

Integration

3.1 Area

Now we turn to a different topic. Area is a quantity that is measured in square units (e.g. square feet). To measure the area of a room that is tiled with 1 foot by 1 foot tiles we just need to count the number of tiles that make up the floor. The total area is the length of the room times its width. When we want to find the area of a region that is not rectangular we still have to resort to some sort of rectangular model to do the computations. This is what we do if we compute the area of a triangle, for example. A triangle with base length b and height h has one half the area of a rectangle with length b and height h . See Figure 3.1.

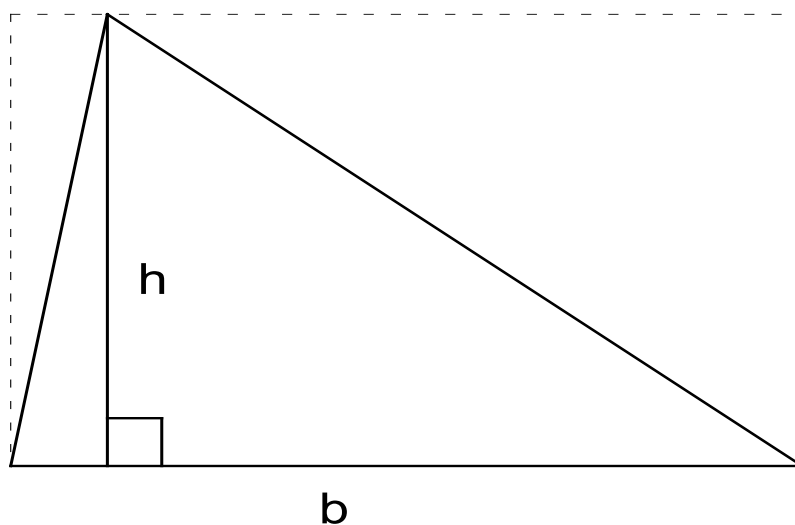


Figure 3.1: Area of a triangle

If we are trying to find the area of a region with a curved boundary we

can again resort to using rectangles as our building blocks. Consider the situation where a region in the coordinate plane is bounded by the x -axis on the bottom, the vertical line $x = a$ on the left, the vertical line $x = b$ on the right, and the graph of some function on the top. Figure 3.2 shows the region bounded by the x -axis below, the vertical line $x = 0$ (the y -axis) on the left, the vertical line $x = 1$ on the right, and the graph of the function given by $f(x) = x^2 + \frac{1}{2}$ on top.

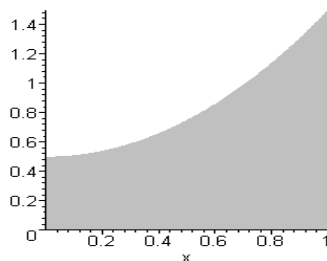


Figure 3.2: Area under the graph of $y = x^2 + \frac{1}{2}$

In problems 135 to 137 we will consider this region. We can find an approximation by using rectangles as Figures 3.3 and 3.4 indicate. In each case there are five rectangles. In one case the sum of the areas of the rectangles overestimates the area of the region, in the other case the sum of the areas of the rectangles underestimates the area of the region.

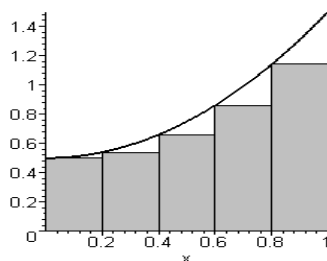


Figure 3.3: Lower rectangles

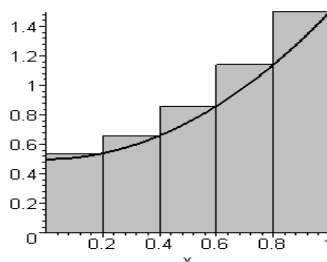


Figure 3.4: Upper rectangles

Problem 135. For each picture above find the total area of the five rectangles shown. Use the formula $f(x) = x^2 + \frac{1}{2}$. Do not rely on the numbers indicated on the axes in the pictures.

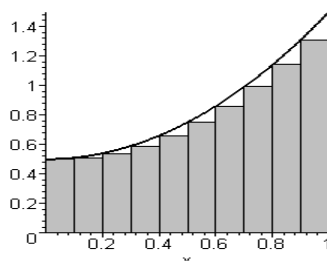


Figure 3.5: Ten lower rectangles

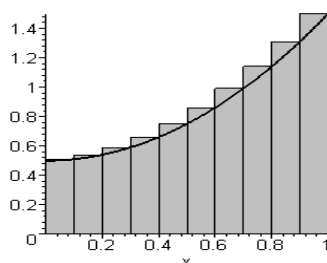


Figure 3.6: Ten upper rectangles

Problem 136. Take the same region referred to in problem 135. Find ten rectangles of equal width whose tops stay below the graph and ten rectangles of equal width whose tops stay above the graph and compute the total area of the rectangles in each case. (See Figures 3.5 and 3.6.)

Problem 137. Note that the area of the region shown in figure 3.2 is always smaller than the sum of areas of rectangles that extend above the graph and always larger than the sum of the areas of rectangles that lie below the graph. How many rectangles would it take to get an area estimate that is within .01 square units of the actual area. (Hint: If the exact area is between any two numbers L and U , where L is smaller than U , then $\frac{U+L}{2}$ is within $\frac{U-L}{2}$ square units of the exact area. Confirm this for yourself. Do this by putting the numbers L , U , and $\frac{U+L}{2}$ on a number line and confirming that any number between L and U is within $\frac{U-L}{2}$ units of $\frac{U+L}{2}$.)

Suppose f is a function that has a domain that includes the closed interval $[a, b]$, (i.e. the numbers x for which $a \leq x \leq b$). A **partition** of $[a, b]$ is a

way of subdividing the interval $[a, b]$ into smaller closed subintervals. More precisely a partition of $[a, b]$ is a choice of numbers (all different) in the closed interval $[a, b]$ such that one of the numbers is a and the other is b . To keep the order of the numbers straight we will use subscripts. The smallest number is equal to a and is called x_0 . The largest number is equal to b and is called x_n where n is the number of subdivisions. The other numbers are given subscripts corresponding to their relative position. In other words $a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$. With this notation we can define a partition by the set of points $\{x_0, x_1, x_2, \dots, x_n\}$. Figure 3.7 shows a partition of $[a, b]$ with 7 subdivisions.



Figure 3.7: Partition

Problem 138. For each of the following a left endpoint a , a right endpoint b , and a number n are given. Find numbers $x_0, x_1, \dots, x_n$ so that $\{x_0, x_1, x_2, \dots, x_n\}$ is a partition of $[a, b]$ into n subintervals of equal width.

1. $a = 0, b = 1$, and $n = 10$.
2. $a = 0, b = 1$, and $n = 20$.
3. $a = 1, b = 2$, and $n = 5$.
4. $a = 2, b = 5$, and $n = 6$.

Problem 139. Consider the function f given by $f(x) = x^2$. For each of the partitions from problem 138, find a number x in each subinterval at which f takes on its largest value.

Suppose f is a function with a domain that contains the closed interval $[a, b]$. Suppose also that $P = \{x_0, x_1, x_2, \dots, x_n\}$ is a partition of that closed interval into n subdivisions. A number x in the i th subdivision at which f takes on its largest value is given the notation x_i^U .

Problem 140. Consider again the function f given by $f(x) = x^2$. Use notation discussed immediately above and the partitions from problem 138 to compute $f(x_1^U)(x_1 - x_0) + f(x_2^U)(x_2 - x_1) + \dots + f(x_n^U)(x_n - x_{n-1})$ for each of the four partitions.

Problem 141. Use pictures to illustrate that the quantities obtained in problem 140 are overestimates of the area under the graph of f , to the right of $x = a$, to the left of $x = b$, and above the x -axis.

Again, let f be a function with a domain that contains the closed interval $[a, b]$ and $P = \{x_0, x_1, x_2, \dots, x_n\}$ be a partition of that closed interval into n subdivisions. A number x in the i th subdivision at which f takes on its smallest value is given the notation x_i^L .

Problem 142. Use the function f and the four partitions from problem 138. Compute $f(x_1^L)(x_1 - x_0) + f(x_2^L)(x_2 - x_1) + \dots + f(x_n^L)(x_n - x_{n-1})$ for each partition and show that each number obtained this way is an underestimate of the area under the graph of f , to the right of $x = a$, to the left of $x = b$, and above the x -axis.

3.2 Riemann Sums

Now we introduce some terminology for the quantities we have been working with. Suppose f is a function with a domain that contains the closed interval $[a, b]$. Suppose also that $P = \{x_0, x_1, x_2, \dots, x_n\}$ is a partition of that closed interval into n subdivisions. An **Upper Riemann Sum** for f , $[a, b]$, and P is a sum of the form $f(x_1^U)(x_1 - x_0) + f(x_2^U)(x_2 - x_1) + \dots + f(x_n^U)(x_n - x_{n-1})$. Here the numbers $x_1^U, x_2^U, x_3^U, \dots, x_{n-1}^U$, and x_n^U are chosen so that x_1^U is in $[x_0, x_1]$ and $f(x_1^U) \geq f(x)$ for all x in $[x_0, x_1]$, x_2^U is in $[x_1, x_2]$ and $f(x_2^U) \geq f(x)$ for all x in $[x_1, x_2]$, and so on. In other words a point x_i^U is chosen in each subinterval $[x_{i-1}, x_i]$ so that the function f takes on its largest value at the point x_i^U . If $f(x) > 0$ for all x in $[a, b]$ then an Upper Riemann Sum is just a fancy way of describing an overestimate by rectangles of the area under the graph of f , to the right of $x = a$, to the left of $x = b$, and above the x -axis. **Lower Riemann Sum** is defined similarly except that the points x_i^L are chosen so that $f(x_i^L)$ is the smallest value of $f(x)$ for all x in $[x_{i-1}, x_i]$. In the case that $f(x) > 0$ for all x in $[a, b]$ the Lower Riemann sum gives us an underestimate by rectangles of the area under the graph of f , to the right of $x = a$, to the left of $x = b$, and above the x -axis.

Problem 143. Let $f(x) = \frac{1}{x}$ for $x > 0$ and consider the closed interval $[1, 2]$ and the partition $P = \{1, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2\}$ of that closed interval. Find an Upper Riemann Sum and a Lower Riemann Sum for P and draw a picture to illustrate the rectangles whose areas are summed up in each Riemann Sum.

Problem 144. Let $f(x) = x$ and consider the closed interval $[0, 1]$ and the partition $P = \{0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}, \frac{n}{n}\}$ of that closed interval into n subintervals of the same width. Find an Upper Riemann Sum and a Lower Riemann Sum for P in terms of n .

Problem 145. Find a formula in terms of n for the sum of the integers from 1 through n . (Hint: If $S(n)$ is the sum of the integers from 1 through n then

$S(n+1) = S(n) + n + 1$ for every positive integer n and $S(2) = 3$. See if you can find a quadratic function S of n for which the relationship $S(n+1) = S(n) + n + 1$ holds for every positive integer n and $S(2) = 3$.)

Problem 146. Use the formula obtained in problem 145 to simplify the expressions obtained in problem 144.

Problem 147. Show that by making n large enough you can make both the Upper Riemann Sum and the Lower Riemann Sum for $f(x) = x$ on $[0, 1]$ with the partition $P = \{0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}, \frac{n}{n}\}$ as close to $\frac{1}{2}$ as you want. How big does n need to be if you want the Upper Riemann Sum to be no bigger than $\frac{1}{2} + 0.0001$?

Problem 148. Find a formula for the sum $1^2 + 2^2 + 3^2 + \dots + (n-1)^2 + n^2$ in terms of n . (Hint: See if you can find a cubic polynomial in the variable n that works. Note that if $S(n)$ represents the sum $1^2 + 2^2 + 3^2 + \dots + (n-1)^2 + n^2$, then $S(n+1) = S(n) + (n+1)^2$ for all positive integers n and $S(2) = 5$. Let $S(n) = An^3 + Bn^2 + Cn + D$ and find A, B, C , and D so that all of the conditions are satisfied.)

Problem 149. Let $f(x) = x^2$ and consider the closed interval $[0, 1]$ together with a general partition P of $[0, 1]$ into n subintervals of equal width. Write down and simplify an expression for an Upper Riemann Sum for f , $[0, 1]$, and P in terms of n .

Problem 150. What number does the expression from problem 149 approach as n increases without bound? This number is the area of the region in the coordinate plane that is bounded by the x -axis on the bottom, the vertical line $x = 0$ on the left, the vertical line $x = 1$ on the right, and the graph of the function given by $f(x) = x^2$ on the top.

3.3 The Area Function

Next we make a definition and introduce some notation for this definition. Suppose we are given a function f and a closed interval $[a, b]$ contained in the domain of f . If f is continuous (for a definition see Appendix A) at each number in $[a, b]$ then there is a single number S that is bigger than or equal to every Lower Riemann Sum for f and $[a, b]$ and smaller than or equal to every Upper Riemann Sum for f and $[a, b]$ no matter which partition is chosen. This number S is called the **definite integral** of f from a to b . The notation for S is $\int_a^b f$.

Suppose f is a function and $[a, b]$ is a closed interval contained in the domain of f on which f is continuous and on which $f(x)$ is never negative. In this case it is not hard to convince yourself that $\int_a^b f$ is the area of the region in the coordinate plane that is bounded by the x -axis on the bottom,

the vertical line $x = a$ on the left, the vertical line $x = b$ on the right, and the graph of f on the top (see figure 8).

In problems 151 to 154 we will assume that f is the function given by $f(x) = x$.

Problem 151. Find the values of $\int_0^1 f$, $\int_0^2 f$, and $\int_0^3 f$ by interpreting these quantities as areas in the sense described above. Draw pictures of the regions whose areas are given by each of the three quantities.

Problem 152. Find the value of $\int_0^a f$ in terms of a .

Problem 153. Problem 152 allows us to define a new function which we will call F . The input is a positive number a and the output is $\int_0^a f$. We can think of this function as an area function in the sense that $F(a)$ gives us the area under the graph of f from the y -axis up to the vertical line $x = a$. Suppose a and b are numbers with $0 < a < b$. Find the value of $F(b) - F(a)$ in terms of a and b and draw a picture of a region whose area is $F(b) - F(a)$.

Problem 154. Suppose F is the function described in problem 153. Find $\lim_{b \rightarrow a} \frac{F(b) - F(a)}{b - a}$. You have found $F'(a)$.

Problem 155. Let $f(x) = x^2$. Let F be the function given by $F(x) = \int_0^x f$ for $x > 0$. Note that $F(a)$ is the area of the region above the x -axis, below the graph of f , to the right of the y -axis, and to the left of the vertical line $x = a$ and that $F(b)$ is the area of the region above the x -axis, below the graph of f , to the right of the y -axis, and to the left of the vertical line $x = b$. Use pictures to show that if $b > a > 0$ then $a^2(b - a) \leq F(b) - F(a) \leq b^2(b - a)$.

Problem 156. Use the inequality from problem 155 (and the Squeeze Theorem) to show that $\lim_{b \rightarrow a^+} \frac{F(b) - F(a)}{b - a} = a^2$. (Note that the notation here indicates that we are taking a right hand limit which is discussed immediately before Theorem 89 and in Appendix A.)

Note that arguments similar to those used in problems 154 and 155 but with $a > b > 0$ will show that $\lim_{b \rightarrow a^-} \frac{F(b) - F(a)}{b - a} = a^2$, and thereby establish that $\lim_{b \rightarrow a} \frac{F(b) - F(a)}{b - a} = a^2$.

Problem 157. Let $f(x) = \frac{1}{x}$. Find an estimate of $\int_1^2 f$ that you are confident is within .05 of the exact value using upper and lower Riemann sums. (Hint: Refer back to problem 137.)

3.4 A Special Area Function

In problem 158 we introduce a function called L . This function will appear in several problems. Make sure you understand how this function is defined. The definition of L is not in terms of a formula. We don't define it by writing $L(x) = \text{some expression in } x$. Instead we define it in terms of a process. We describe a way in which one might, at least in theory, obtain values for this function. This is akin to the unit circle definition of the sine function which goes something like this: "Given a positive number t , start at the point $(1, 0)$ on the unit circle, move a distance of t units along the circle in the counterclockwise direction, $\sin(t)$ is the y -coordinate of the point at which you end up. If $t < 0$ you move $|t|$ units in a clockwise direction and again take $\sin(t)$ to be the y -coordinate of the point at which you end up, $\sin(0)$ is just taken to be 0."

For $t > 1$, $L(t)$ is the area under the graph of $y = \frac{1}{x}$, above the x -axis, to the right of the line $x = 1$, and to the left of the line $x = t$. For $0 < t < 1$ $L(t)$ is the negative of the area under the graph of $y = \frac{1}{x}$, above the x -axis, to the right of the line $x = t$, and to the left of the line $x = 1$. $L(1)$ is just taken to be 0. (See Figures 3.8, 3.9, and 3.10 below.) The beauty of each of the functions $\sin$ and L is not the definition, but the incredible range of problems that each can shed light on.

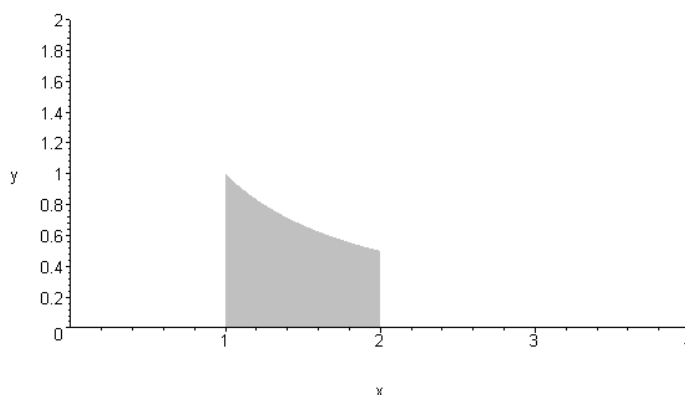
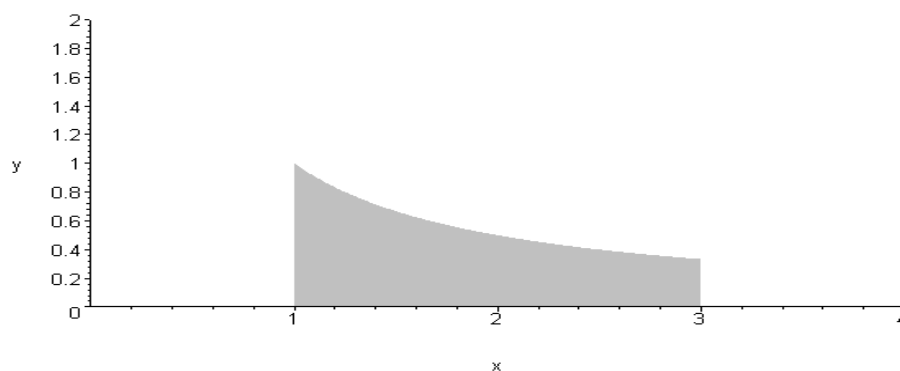
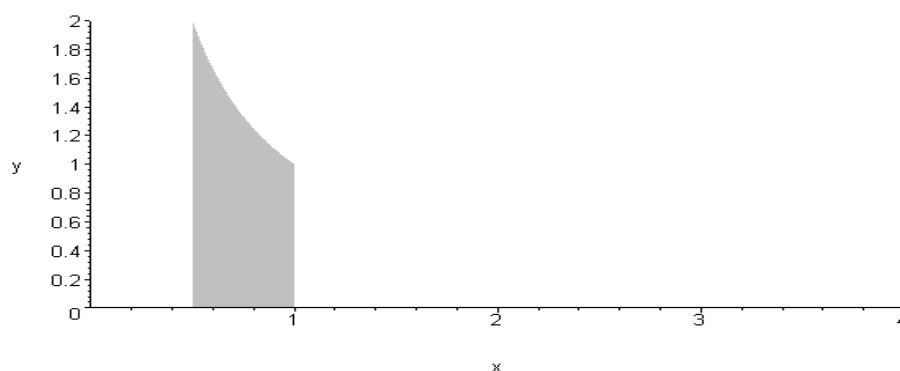


Figure 3.8: $L(2)$

Note If $b < a$ we use the notation $\int_a^b f$ to stand for $(-\int_b^a f)$. The notation $\int_a^a f$ is defined to be the number 0.

Figure 3.9: $L(3)$ Figure 3.10: $L(1/2)$

Problem 158. Let $f(x) = \frac{1}{x}$. By the previous note the function L can be defined for $x > 0$ by $L(x) = \int_1^x f$. Use pictures to show that for $b > 2$,

$$(b-2)\frac{1}{b} \leq L(b) - L(2) \leq (b-2)\frac{1}{2}$$

and that for $1 < b < 2$,

$$\frac{1}{2}(2-b) \leq L(2) - L(b) \leq \frac{1}{b}(2-b).$$

Problem 159. Use problem 158 and the squeeze theorem for right and left handed limits to show that $L'(2) = \frac{1}{2}$.

Problem 160. Using problems 158 and 159 as a model show that for any $a > 0$ $L'(a) = \frac{1}{a}$. (You will need to look at pictures with $a < 1$ and $a > 1$ separately.)

3.5 The Fundamental Theorem of Calculus

Theorem 161 (The Vertical Shift Theorem). In what follows we will use the following geometrically obvious fact (see Figure 3.11 below): If two functions have derivatives that are defined and equal for every number in some interval (their tangent lines have the same slope at every value of x) then the graph of one function is just a vertical shift of the graph of the other. In other words if I is an interval and $f'(x) = g'(x)$ for every number in I (and both derivatives are defined for every number in I) then $f(x) = g(x) + C$ for all x in I and some constant C . (This fact follows from a theorem called the Mean Value Theorem which we will not prove in this class.)

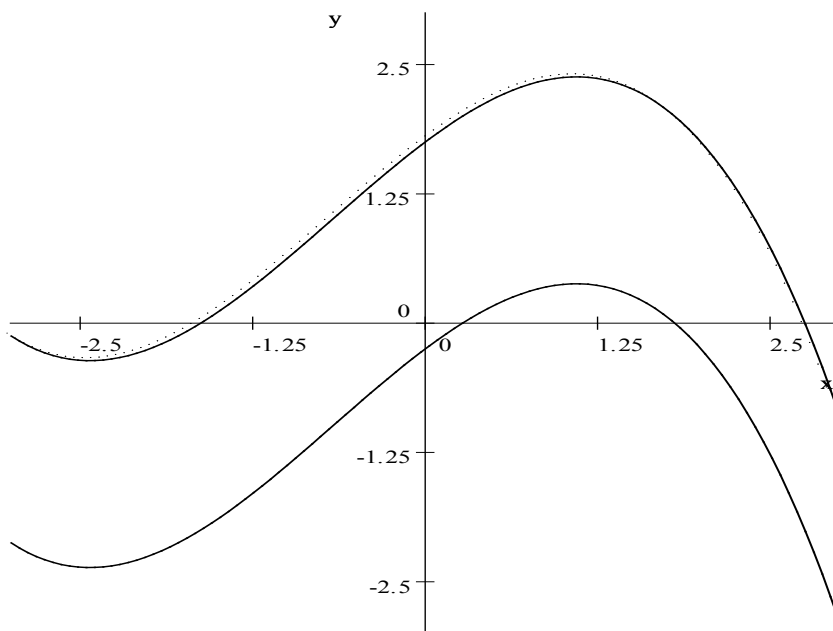


Figure 3.11: Vertical Shift

Problem 162. Let $f(x) = x^3$. Define a function G for all t by $G(t) = \int_0^t f$. Use pictures and the squeeze theorem to show that $G'(t) = t^3$. (See problems 155 and 156.)

Problem 163. Again let $f(x) = x^3$. Use the result from problem 162, namely that $G'(t) = t^3$, to come up with another formula for G and to obtain the numerical value of $\int_1^2 f$.

Problem 164. Using ideas similar to those from problems 162 and 163 find the values of:

1. $\int_1^2 f$ where $f(x) = x^4$.
2. $\int_0^\pi f$ where $f(x) = \sin x$
3. $\int_0^1 f$ where $f(x) = \frac{1}{1+x^2}$. (Hint: If $G(x) = \arctan x$ then $G'(x) = \frac{1}{1+x^2}$.)

Problems 162 to 164 illustrate the following important theorem:

Theorem 165 (The Fundamental Theorem of Calculus). Suppose we are given a function f and a closed interval $[a, b]$ that is contained in the domain of f . Suppose also that f is continuous (see Appendix A) at every number in $[a, b]$. Then the following are all true.

1. The function G defined by $G(t) = \int_a^t f$ is also continuous at each number in $[a, b]$.
2. For each number x in (a, b) $G'(x) = f(x)$.
3. If c and d are numbers in $[a, b]$, and F is any function with the property that $F'(x) = f(x)$ for every x in $[a, b]$, (F is an **antiderivative** of f on $[a, b]$), then $\int_c^d f = F(d) - F(c)$.

Problem 166. Use the Fundamental Theorem of Calculus to find the values of each of the following definite integrals. All you need to do is to find an antiderivative and then use part three of the Fundamental Theorem of Calculus.

1. $\int_2^5 f$ where $f(x) = (3x^2 + 2x + 1)$
2. $\int_0^{\pi/2} f$ where $f(x) = \cos x$
3. $\int_1^9 f$ where $f(x) = \sqrt{x}$
4. $\int_0^1 f$ where $f(x) = (x+1)(x-2)$
5. $\int_0^1 f$ where $f(x) = (x+1)^5$
6. $\int_1^2 f$ where $f(x) = \frac{1}{x^2}$

3.6 Properties of L

Recall that we defined the function L for all positive numbers t by $L(t) = \int_1^t f$ where $f(x) = \frac{1}{x}$. The function L has two basic properties. They are that $L(1) = 0$ and that $L'(x) = \frac{1}{x}$. From these two properties we can establish a number of other properties that L has. They are developed in problems 167 to 171.

The properties of L that will be established are:

1. For all positive numbers a and b $L(ab) = L(a) + L(b)$.
2. For all positive numbers a and rational numbers r $L(a^r) = rL(a)$.

Problem 167. Let a be a positive number and define the function Q for $x > 0$ by $Q(x) = L(ax)$. Use the chain rule and the formula for the derivative of L to show that for all $x > 0$, $Q'(x) = L'(x)$. This shows us (by the Vertical Shift Theorem) that $Q(x) = L(x) + C$ for some constant C and all $x > 0$.

Problem 168. Letting $Q(x) = L(ax)$ as in problem 167, find an expression for the constant C in terms of L by plugging 1 in for x in the expression $Q(x) = L(x) + C$ and noting that $L(1) = 0$ and $Q(1) = L(a)$.

Problem 169. Put problems 167 and 168 together to show that for any numbers $a > 0$ and $x > 0$ $L(ax) = L(a) + L(x)$.

Problem 170. Show that for any rational number r and any real number x , $L(x^r) = rL(x)$. (Hint: Define a new function P by $P(x) = L(x^r)$. Use the chain rule to show that $P'(x) = rL'(x)$.)

Problem 171. Show that L is increasing and concave down for all positive real numbers.

Problem 172. Sketch a rough graph of L using the fact that $L(1) = 0$.

Recall that a f function is one to one if whenever x_1 and x_2 are in the domain of f , and $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. In other words two different inputs cannot yield the same output. Every increasing and every decreasing function is one to one.

Problem 173. By problem 171 L is one to one and thus has an inverse. Let us give the inverse of L the name E . What is $E(0)$?

Problem 174. What is $E'(x)$? (Hint: Since E and L are inverses $L(E(x)) = x$ is an identity. Now differentiate both sides of the identity using the fact that $L'(x) = \frac{1}{x}$.)

Problem 175. Show that the function f given by the formula $f(x) = C \cdot E(rx)$ satisfies the identity $f'(x) = rf(x)$.

3.7 More Motion

Problem 175 gives us a family of functions that can be used to describe any situation where the instantaneous rate of change of a quantity is proportional to the quantity itself. In problems 176 and 177 we investigate some of these situations.

We start by looking at a situation where frictional forces are proportional and opposite in direction to velocity. In the special case where friction is the only force acting on an object in motion, Newton's Second Law of Motion gives us that mass times acceleration is equal to a negative constant times velocity. In symbols this is $ma = -kv$ where m is mass, a is acceleration, v is velocity, and k is some positive number. In the past we have used $f(x)$ to stand for position at time x . This time we will use $f(x)$ to stand for velocity at time x . This means that $f'(x)$ is acceleration at time x . Now we can translate $ma = -kv$ into $mf'(x) = -kf(x)$ or $f'(x) = \frac{-k}{m}f(x)$ and use problem 175 to find f .

Problem 176. At an ice show a large statue of Frosty the Snowman on ice skates is given a push and allowed to glide in a straight line along the ice without any other forces except friction acting on it. Suppose that Frosty's velocity at time x seconds after he was given the initial push is $f(x)$ meters per second. Since Frosty is moving horizontally the force of gravity does not influence Frosty's motion. The only force exerting any significant influence on his motion is the force of friction. Suppose that this frictional force is directly proportional and opposite in sign to Frosty's velocity. Suppose Frosty has a mass of 80 kg, his initial velocity is 15 m/s, and the constant of proportionality is -20 kg/s. Use problem 175 and the discussion immediately above to find a formula for the function f that gives Frosty's velocity in terms of the function E introduced in problem 173.

Problem 177. Find a formula in terms of E for Frosty's position function. (Hint: Recall that if g is the position function then $g' = f$ from problem 176. We might as well let $g(0) = 0$. Now use the fact that $E' = E$, the chain rule, and good old fashioned guess and check to come up with a formula for g in terms of E .)

In problems 178 and 179 you will prove some properties of the function E . They are:

1. $E(x+y) = E(x)E(y)$ for any real numbers x and y .
2. $E(rx) = (E(x))^r$ for any real number x and any rational number r .

Along with the properties of the function L proved in problems 169 and 170 you will use the fact that $L(E(x)) = x$ is an identity and the fact that if $L(a) = L(b)$ then $a = b$ (L is one to one).

Problem 178. Use the fact that $L(ab) = L(a) + L(b)$ for all positive numbers a and b to prove that $E(x)E(y) = E(x+y)$ for all real numbers x and y .

Problem 179. Use the fact that $L(a^r) = rL(a)$ for all rational numbers r and positive numbers a to prove that $(E(x))^r = E(rx)$ for all real numbers x and rational numbers r .

Problem 180. Show that for every rational number r , $E(r) = (E(1))^r$.

Because of the importance of the functions E and L in both practical and theoretical work, algorithms for approximating values of these functions are built into most calculators and most mathematical software. The function E is commonly called the exponential function because of the properties established in problems 178, 179, and 180. Standard notation for the value of the function at the number x is either $\exp(x)$ or e^x . The letter e stands for the number $\exp(1)$. The number e is an irrational number like π . A fifty decimal place approximation of e is:

2.71828182845904523536028747135266249775724709369996

The function L is commonly called the natural logarithm function and the standard notation for $L(x)$ is $\ln(x)$. From a calculus standpoint the fact that these two functions act like exponential and logarithmic functions is not nearly as important as the fact the exponential function is its own derivative and that the derivative of the natural logarithm function is the function f given by $f(x) = \frac{1}{x}$, $x > 0$.

For list of properties of the natural logarithm and the exponential function see appendix B.

Problem 181. Use the result from problem 177 and the properties of the natural logarithm and exponential functions to find the approximate total distance that Frosty travels. (Hint: This would be the value that Frosty's position function approaches for arbitrarily large positive time values.)

At the beginning of the semester we looked at the motion of a ball. We assumed that the only force acting on the ball was gravitational and that this produced a constant acceleration (or deceleration depending on how you look at it). A more realistic model of motion includes frictional forces that act opposite to the direction of the motion. In problems 182 to 185 we will look at the motion of the ball again, this time taking into account frictional forces. Because the ball is on a sloped ramp, the component of gravitational force acting in line with the ball is a small fraction of the force that is exerted in the vertical direction (see Figure 3.12).

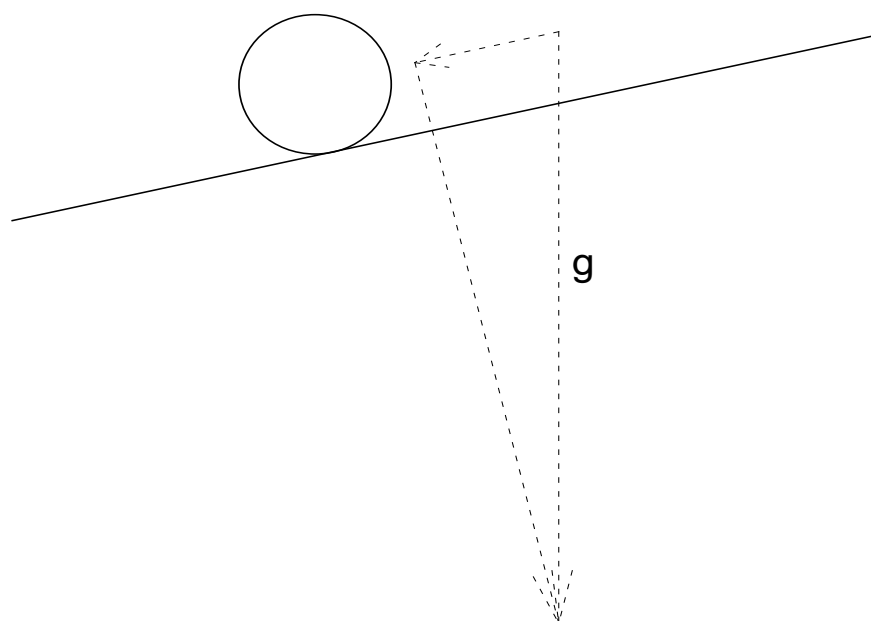


Figure 3.12: Force Diagram

In order to keep the numbers as simple as possible we will slope the ramp in such a way that this force in newtons is numerically equal to the mass in kilograms of the ball. In the absence of friction this would produce a constant acceleration of -1 m/s. By including a frictional force that is proportional in magnitude and opposite in direction to the velocity we get from Newton's second law of motion that $\text{mass} \cdot \text{acceleration} = \text{gravitational force} + \text{frictional force}$. Let v be the velocity function and a the acceleration function. Using meters for distance and seconds for time, Newton's second law of motion translates, in our situation, to the equation:

$$m \cdot a(x) = -m - kv(x) \text{ or}$$

$$(\text{EQ. 1}) \quad a(x) = -1 - \frac{k}{m}v(x)$$

Here m stands for mass and k is the proportionality constant that relates

the frictional force to the velocity. If we differentiate both sides of EQ. 1 we get

$$a'(x) = -\frac{k}{m}v'(x).$$

Using the fact that $v' = a$ we obtain

$$\text{(EQ. 2) } a'(x) = -\frac{k}{m}a(x).$$

Problem 182. Assume for simplicity that $\frac{k}{m} = 0.1$. Use item 16 from the properties of the natural logarithm and exponential functions and EQ. 2 to find a general formula for the function a that satisfies the equation immediately above. (Remember to include a constant C in order to keep your answer as general as possible for the work that follows.)

Problem 183. Use problem 182 and equation EQ. 1 above to find a formula for the corresponding function v that also satisfies the equation $v(0) = 2$. (In other words the initial velocity is 2 m/s.)

Problem 184. Let f be the position function for the motion of the ball described in problems 182 and 183 and the discussion that preceded them. Assume that $f(0) = 0$. Find a formula for f .

Problem 185. Sketch a graph of the function f from problem 184.

Problem 186. Find formulas for the derivatives of the functions whose formulas are given:

1. $f(x) = \ln(x^2 + 1)$
2. $f(x) = \ln(-x)$ where $x < 0$
3. $f(x) = \exp(-x^2)$
4. $f(x) = x \exp(-x)$
5. $f(x) = \ln(\sec x)$ where $-\frac{\pi}{2} < x < \frac{\pi}{2}$
6. $f(x) = \ln(\sin x)$ where $0 < x < \pi$
7. $f(x) = \ln(\sec x + \tan x)$ where $-\frac{\pi}{2} < x < \frac{\pi}{2}$
8. $f(x) = x \ln x - x$ where $x > 0$
9. $f(x) = \exp(-x) \sin x$
10. $f(x) = \frac{\exp(x) - \exp(-x)}{2}$

3.8 Antiderivatives

Antiderivatives allow us to work backwards from acceleration to velocity or from velocity to position. Rate information or relationships involving rates are often easier to obtain than information about the underlying quantity. Antiderivatives give us one tool for working backwards from rate information. We can also use antiderivatives when we evaluate definite integrals using the Fundamental Theorem of Calculus.

The process of starting with a formula for a function and coming up with a formula for an antiderivative is not quite as straightforward as taking the derivative of a function. Some derivative rules just aren't as easy to run in reverse. We start by creating a table of derivative rules.

Problem 187. *Create a comprehensive table of derivatives. In the left hand column you should include all the elementary functions that you know (e.g. the trig functions, the inverse trig functions, the exponential function, and the natural log function). You should also include expressions obtained by doing arithmetic and composition to elementary functions, for example $f(x)g(x)$, $cf(x)$, $f(x) + g(x)$, or $f(g(x))$. For these latter expressions you can write the corresponding derivatives in terms of the original functions and their derivatives. For example the right-hand entry that corresponds to the left hand entry $f(g(x))$ should read $f'(g(x))g'(x)$.*

A table of antiderivatives could be constructed by just swapping the columns in the table you just created. Notice, however, that some of the expressions on the right-hand side of the derivative table are quite a bit more complicated than their counterparts on the left and that some simple expressions are in fact missing from the right-hand column of your derivative table. For example you don't have a simple composition or a simple product in your right-hand column.

Problem 188. *Create an antiderivative table by swapping the columns of your derivative table.*

Techniques for finding antiderivatives will be covered extensively in the second semester of calculus. In addition to the straightforward applications of the table you just created we will investigate one special technique. It involves running the chain rule backwards.

Notice that any expression of the form $f(g(x))g'(x)$ is just the result of an application of the chain rule. This means that we have a way of obtaining an antiderivative of such an expression. To see this, suppose that F is an antiderivative of f . Then $F(g(x))$ is an antiderivative of $f(g(x))g'(x)$ since

the chain rule tells us that the derivative of $F(g(x))$ is $F'(g(x))g'(x)$ which is precisely $f(g(x))g'(x)$ by virtue of the fact that $F' = f$. It is as if we simply took the right-hand entry $F'(g(x))g'(x)$ of the derivative table and looked at its left-hand counterpart $F(g(x))$. The only difference is that what we now call F' is called f in the original table.

If we can turn a given expression into the form $K \cdot f(g(x))g'(x)$ where K is some constant then we have a way of finding an antiderivative. For example consider the function given by $f(x) = x(x^2 + 1)^3$. By rewriting this function in the form $\frac{1}{2} \cdot (x^2 + 1)^3 \cdot 2x$ we have $\frac{1}{2}f(g(x))g'(x)$ where $f(x) = x^3$ and $g(x) = x^2 + 1$. Since $F(x) = \frac{x^4}{4}$ is the formula for an antiderivative of f we get that $\frac{1}{2}F(g(x)) = \frac{1}{2} \cdot \frac{(x^2 + 1)^4}{4}$ is an antiderivative of $x(x^2 + 1)^3$. All you need to do to check whether you have the right antiderivative is to take the derivative and see if you get what you started with. Try it and see.

Problem 189. Find formulas for the most general antiderivatives of the functions whose formulas are given:

1. $f(x) = \frac{1}{x}, x > 0$
2. $f(x) = \frac{1}{x}, x < 0$
3. $f(x) = \frac{1}{x+4}, x > -4$
4. $f(x) = \frac{1}{2x-3}, x > \frac{3}{2}$
5. $f(x) = \frac{1}{ax+b}, x > -\frac{b}{a}$
6. $f(x) = x(x^2 + 1)^5$
7. $f(x) = x\sqrt{x^2 + 1}$
8. $f(x) = x \sin(x^2)$
9. $f(x) = \exp(rx)$
10. $f(x) = x \exp(-x^2)$
11. $f(x) = (1 + \sin x)^5 \cos x$
12. $f(x) = \frac{x}{1+x^2}$

13. $f(x) = \frac{\ln x}{x}, x > 0$

14. $f(x) = \tan x, -\frac{\pi}{2} < x < \frac{\pi}{2}$ (Hint: Use Problem 186 part 5.)

15. $f(x) = \cot x, 0 < x < \pi$ (Hint: Use Problem 186 part 6.)

16. $f(x) = \sec x, -\frac{\pi}{2} < x < \frac{\pi}{2}$ (Hint: Use Problem 186 part 7.)

Chapter 4

Appendix A: Background on Limits and Continuity

Before we get into the technical details we will see whether our intuition serves us in determining various limits. To that end, see what you can come up with in the following problem.

Problem 190. *Make a guess at the following limits. Explain your reasoning.*

1. $\lim_{x \rightarrow 3} (5x + 3)$
2. $\lim_{x \rightarrow 2} (x^2 + 1)$
3. $\lim_{x \rightarrow 0} (\sqrt{x+3})$
4. $\lim_{x \rightarrow 7} \left(\frac{1}{x}\right)$
5. $\lim_{x \rightarrow 3} |x - 3|$
6. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$
7. $\lim_{x \rightarrow 1} (ax^3 + bx^2 + cx + d)$
8. $\lim_{x \rightarrow 0} \begin{cases} x + 1 & \text{if } x \leq 0 \\ 1 - 2x & \text{if } x > 0 \end{cases}$
9. $\lim_{x \rightarrow 0} \begin{cases} 2x & \text{if } x < 0 \\ 5 & \text{if } x = 0 \\ x^2 & \text{if } x > 0 \end{cases}$
10. $\lim_{x \rightarrow 2} \begin{cases} x + 1 & \text{if } x \leq 2 \\ 1 - 2x & \text{if } x > 2 \end{cases}$

$$11. \lim_{x \rightarrow 1} \left(\frac{\sqrt{x} - 1}{x - 1} \right)$$

4.1 Precise Definition of Limit

When we say that $\lim_{x \rightarrow a} f(x) = L$ we are saying that the quantity $f(x)$ can be made arbitrarily close to the fixed number L just by making x sufficiently close to a . The way to make this a little more precise is to say that if someone were to give you a very small positive number ε (the Greek letter epsilon) and tell you that they wanted you to make $f(x)$ within ε units of L , then you could find a positive number δ (the Greek letter delta) so that if x was within δ units of a then $f(x)$ would be within ε of L . Another way to say this is that corresponding to every small positive number ε that is given to you there must be a δ that works, no matter how small ε might happen to be.

Definition 191. We say that $\lim_{x \rightarrow a} f(x) = L$ if it is true that whenever a positive number ε is specified, there is a corresponding positive number δ with the property that whenever x is within δ units of a , that is whenever $a - \delta < x < a + \delta$, then $f(x)$ is within ε units of L , that is $L - \varepsilon < f(x) < L + \varepsilon$.

To illustrate this definition we will use it to confirm that $\lim_{x \rightarrow 4} (2x + 1) = 9$ as expected.

Note that if we want $2x + 1$ to be within ε of 9 we are asking that $2x + 1$ be in between $9 - \varepsilon$ and $9 + \varepsilon$ or that $9 - \varepsilon < 2x + 1 < 9 + \varepsilon$. We can turn this in to an equivalent inequality by first subtracting 1 from each part of the inequality and obtaining $8 - \varepsilon < 2x < 8 + \varepsilon$ and then by dividing each part by 2 to get $4 - \frac{\varepsilon}{2} < x < 4 + \frac{\varepsilon}{2}$. Evidently if $4 - \frac{\varepsilon}{2} < x < 4 + \frac{\varepsilon}{2}$, or in other words if x is within $\frac{\varepsilon}{2}$ of 4, then $2x + 1$ is within ε of 9. In other words, if we want $2x + 1$ to be within ε units of 9 this can be accomplished by making x within $\frac{\varepsilon}{2}$ of 4. The number $\frac{\varepsilon}{2}$ is a number that we can use in place of the δ from the previous paragraph.

4.2 Properties of Limits

The following is a list of properties that can be derived from the precise mathematical definition of limit. These properties should seem intuitively reasonable as well. In the following a, c, L , and M are numbers, and f, g , and h are functions that are defined in an open interval containing a but not

necessarily at a . Note that while these properties will let you determine the value of a limit in certain cases, the properties do not tell the whole story. In particular they do not allow you to prove that a limit does not exist. See “When Limits Don’t Exist” below.

1. The value of $\lim_{x \rightarrow a} f(x)$ does not depend on whether $f(a)$ is defined. Even if $f(a)$ is defined, it need not be the same number as $\lim_{x \rightarrow a} f(x)$.
2. If $f(x) = g(x)$ for all numbers x in an open interval containing a except maybe at $x = a$ then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$.
3. $\lim_{x \rightarrow a} c = c$
4. $\lim_{x \rightarrow a} x = a$
5. Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ then:
 - (a) $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$
 - (b) $\lim_{x \rightarrow a} cf(x) = cL$
 - (c) $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = LM$
 - (d) $\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{L}{M}$ as long as $M \neq 0$
 - (e) $\lim_{x \rightarrow a} \sqrt{f(x)} = \sqrt{L}$ as long as $L > 0$.
6. If $f(x) \leq g(x)$ for all x in an open interval containing a but not necessarily at a and $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ then $L \leq M$.
7. If $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing a but not necessarily at a and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} g(x) = L$ as well. (This is sometimes called the Squeeze Theorem or the Sandwich Theorem.)
8. Right and left hand limits obey the same properties listed above except that you replace “in an open interval containing a except maybe at $x = a$ ” by “in an open interval with left endpoint a ” for right hand limits and by “in an open interval with right endpoint a ” for left hand limits.
9. It is a fact that $\lim_{x \rightarrow a} f(x) = L$ if and only if both $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$ and sometimes the only way to get at $\lim_{x \rightarrow a} f(x) = L$ is to look at $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ separately.

Problem 192. Use the properties of limits to determine the following limits. Indicate which property or properties you are using.

$$1. \lim_{x \rightarrow 2} \left(\frac{x^2 + x - 6}{x - 2} \right)$$

$$2. \lim_{x \rightarrow -1} \left(\frac{\sqrt{x^2 + 1}}{x + 2} \right)$$

$$3. \lim_{x \rightarrow 2} \left(\frac{\frac{1}{x} - \frac{1}{2}}{x - 2} \right)$$

$$4. \lim_{x \rightarrow 0} \left(\frac{x^2}{x + 1} \right)$$

$$5. \lim_{x \rightarrow 0} \left(x^2 \sin \left(\frac{1}{x} \right) \right)$$

4.3 When Limits Don't Exist

Consider the piecewise defined function given by $f(x) = \begin{cases} 1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$.

What if we were asked to determine $\lim_{x \rightarrow 0} f(x)$? Any number L we choose is a least 1 unit away from one or the other of the numbers 1 or -1 . This means that there is no positive number δ for which it is the case that if x is within δ units of 0 (i.e. $-\delta < x < \delta$) then $f(x)$ is less than 1 unit away from whatever L we might have chosen for our limit. Any interval of the form $(-\delta, \delta)$ will contain numbers x for which $f(x) = 1$ and numbers x for which $f(x) = -1$. L can't be both less than 1 unit away from -1 and simultaneously less than 1 unit away from 1. Try it and see. For this function we say that $\lim_{x \rightarrow 0} f(x)$ does not exist. Now let's consider the function given

by $f(x) = \frac{1}{x}$. What can we say about $\lim_{x \rightarrow 0} f(x)$? It is not hard to convince yourself that any interval of the form $(-\delta, \delta)$ contains both numbers larger than 1 and less than -1 . Again we say that $\lim_{x \rightarrow 0} f(x)$ does not exist. Now

let's consider the function given by $f(x) = \frac{1}{x^2}$ and look at intervals of the form $(-\delta, \delta)$. No matter which number L you choose it is always possible to find a number a in $(-\delta, \delta)$ for which it is true that $f(a)$ is larger than $L + 1$. Try it and see.

These are just a few examples to illustrate the fact that $\lim_{x \rightarrow a} f(x)$ does not have to exist. If $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ does not exist for some number a then we say that f is not differentiable at a .

Problem 193. *Show that the absolute value function is not differentiable at 0.*

Problem 194. *Show that the cube root function is not differentiable at 0.*

4.4 Continuity

Our main concern with continuity at the calculus level is that in many cases we need our functions to be "well behaved" in the sense that their graphs are unmarred by holes, breaks, or other aberrant behavior. Part of ensuring that a function has a nice graph is to consider intervals that avoid numbers for which the function is not defined (e.g. zeroes of the denominator). Let's consider a function f that is defined at every number in the closed interval $[a, b]$. When we say f is continuous on $[a, b]$ we mean that for any number c that is strictly between a and b $\lim_{x \rightarrow c} f(x)$ exists and is equal to $f(c)$, that $\lim_{x \rightarrow a+} f(x)$ exists and is equal to $f(a)$, and that $\lim_{x \rightarrow b-} f(x)$ exists and is equal to $f(b)$.

Most functions whose formulas are written down using arithmetic and the elementary functions to which we have been introduced (e.g. the exponential and trigonometric functions and their inverses) are continuous on every closed interval contained in their domain. In order to come up with formulas for functions that are not continuous on some closed interval contained in their domains we are almost forced to resort to piecewise defined notation or functions such as the absolute value function or the greatest integer function which are essentially piecewise defined as well.

If a function fails to be continuous on some closed interval, the reason could be that some limit is not defined or that the limit is defined but not equal to the value of the function. It is the first situation that is of primary interest to us since in the second situation recovering a continuous function is just a matter of redefining the function at a few isolated points.

Consider for example the function f given by $f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ -1 & \text{if } x = 0 \end{cases}$.

If we compute $\lim_{x \rightarrow 0} f(x)$ we get 1. The function f is identical to the function g defined by $g(x) = 1$ with the single exception that the two functions have

different values when $x = 0$. For most applications there would be no harm in just using the function g instead of f .

The function h given by $h(x) = \begin{cases} 1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$, however, permits no such quick fix. There is no way to make h into a function that is continuous on any closed interval containing 0 without changing its value at infinitely many numbers. Here the problem is that $\lim_{x \rightarrow 0} h(x)$ is not defined.

For our purposes when we require that a function f is defined and continuous on some closed interval $[a, b]$ we are looking for an interval in the domain of f where the right hand limit is defined at a , the left hand limit is defined at b , and the two sided limit is defined at every other number in $[a, b]$.

Note that a number c where either $\lim_{x \rightarrow c} f(x)$ fails to exist or where $\lim_{x \rightarrow c} f(x) \neq f(c)$ is called a discontinuity of f . In most calculus books this notion of discontinuity is extended to included numbers missing from the domain, so that for example the number 2 is called a discontinuity of the function given by $f(x) = \frac{1}{x-2}$. I prefer to keep this concept separate from continuity.

Chapter 5

Appendix B: Properties of the natural logarithm and the exponential function

1. The domain of $\ln$ is the set of all positive real numbers and the range of $\ln$ is the set of all real numbers.
2. The domain of $\exp$ is the set of all real numbers and the range of $\exp$ is the set of all positive real numbers.
3. If $f(x) = \exp(x)$ then $f'(x) = \exp(x)$.
4. If $f(x) = \exp(x)$ then the function F given by $F(x) = \exp(x) + C$ is an antiderivative of f for every constant C .
5. If $f(x) = \ln(x)$ and $x > 0$ then $f'(x) = \frac{1}{x}$.
6. If $f(x) = \frac{1}{x}$ and $x > 0$ then the function F given by $F(x) = \ln(x) + C$ is an antiderivative of f for every constant C .
7. $\ln(ab) = \ln(a) + \ln(b)$ for all positive real numbers a and b .
8. $\ln(a^r) = r \ln a$ for all positive real numbers a and rational numbers r .
9. $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$ for all positive real numbers a and $b \neq 0$. (Note that this follows from properties 7 and 8 if we use ab^{-1} in place of $\frac{a}{b}$.)
10. The graph of $\ln$ is concave down and increasing for every number in the domain of $\ln$.
11. The graph of $\ln$ contains the point $(1, 0)$.
12. The y -axis is a vertical asymptote for the graph of $\ln$. That is, $\ln(x)$ takes on arbitrarily large negative values as x takes on positive values close to 0. Note for example that $\ln(e^{-2000}) = -2000$.

13. The graph of $\exp$ is increasing and concave up for all real numbers.
14. The graph of $\exp$ contains the point $(0, 1)$.
15. The x -axis is a horizontal asymptote for the graph of $\exp$. That is, the exponential function takes on positive values arbitrarily close to 0 as x takes on large negative values. Note, for example, that $0 < \exp(-2000) < 10^{-800}$.
16. If C is any real number then the function given by $f(x) = C \exp(rx)$ satisfies the equation $f'(x) = rf(x)$ for all real numbers x .